

Automated Skin Cancer Detection Using Deep Learning (CNN)

AIDS-456: Major Project Dissertation

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for the award of the degree of

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in

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Submitted by

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Greater Noida

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DECLARATION

This is to certify that the material embodied in this Major Project - Dissertation titled "Automated Skin Cancer Detection Using Deep Learning (CNN)" being submitted in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Artificial Intelligence and Data Science is based on my original work. It is further certified that this Major Project - Dissertation work has not been submitted in full or in part to this university or any other university for the award of any other degree or diploma. My indebtedness to other works has been duly acknowledged at the relevant places.

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CERTIFICATE

This is to certify that the work embodied in this Major Project - Dissertation titled "Automated Skin Cancer Detection Using Deep Learning (CNN)" being submitted in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Artificial Intelligence and Data Science, is original and has been carried out by CHETAN SAINI (Enrollment No.- 00525511922) under my supervision and guidance.

It is further certified that this Major Project - Dissertation work has not been submitted in full or in part to this university or any other university for the award of any other degree or diploma to the best of my knowledge and belief.

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ABSTRACT

Skin cancer represents one of the most prevalent and potentially fatal malignancies globally, yet clinical outcomes are heavily dependent on early and accurate diagnosis. This project presents the development of an automated skin lesion classification system utilizing Deep Learning (DL) architectures to provide a reliable decision-support tool for dermatological screening. The proposed system leverages the HAM10000 dataset, a comprehensive collection of 10,015 multi-source dermatoscopic images comprising seven distinct diagnostic categories: Melanocytic nevi (nv), Melanoma (mel), Benign keratosis-like lesions (bkl), Basal cell carcinoma (bcc), Actinic keratoses (akiec), Vascular lesions (vasc), and Dermatofibroma (df).

To overcome the challenges of high class imbalance and limited computational resources, the methodology adopts a Transfer Learning strategy using the MobileNetV2 architecture. The model is optimized via Focal Loss, a specialized loss function designed to prioritize difficult-to-classify samples and minority classes, thereby improving the sensitivity of the model toward malignant lesions. Images are preprocessed to a standard 224×224 pixel resolution and normalized to ensure consistent feature extraction.

The experimental results demonstrate a robust performance with a final test accuracy of 74.8% and a weighted F1-score of 0.72. Beyond the model training, the project includes the deployment of a full-stack web application developed using React.js, Node.js, and MongoDB, allowing users to upload dermatoscopic images and receive instantaneous probability-based predictions. This integration of a high-performance CNN with an accessible interface showcases the potential of AI-driven tools to bridge the gap in healthcare accessibility, particularly in regions with limited access to specialized dermatological care.

Keywords: Deep Learning, Convolutional Neural Networks (CNN), Skin Cancer Detection, MobileNetV2, HAM10000, Transfer Learning, Focal Loss, Medical Imaging.

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CHAPTER 1: INTRODUCTION

The global healthcare landscape is currently confronting an unprecedented escalation in the incidence of dermatological malignancies, a phenomenon that has transformed skin cancer into a critical public health priority across every inhabited continent. As the largest organ of the human body, the skin serves as the primary interface between the internal biological system and the external environment, rendering it uniquely susceptible to a diverse array of pathologies triggered by ultraviolet (UV) radiation, chemical pollutants, and genetic predispositions. Recent epidemiological data from the World Health Organization (WHO) and the Global Cancer Observatory (GLOBOCAN) indicate that skin cancer now ranks as the 17th most common cancer worldwide, with over 1.2 million new cases of non-melanoma skin cancer and approximately 331,722 new cases of melanoma reported in 2022 alone. The trajectory of these statistics is particularly alarming in high-UV regions; for instance, in Australia and New Zealand, the age-standardized incidence rates exceed 30 cases per 100,000 individuals, reflecting an environment where intense solar exposure significantly accelerates malignant transformations in the epidermal layers.

The clinical urgency of this crisis is underscored by the stark disparity in survival outcomes between early-stage and advanced-stage diagnoses. Melanoma, while accounting for only about 1% of total skin cancer cases, is responsible for the vast majority of skin cancer-related fatalities due to its high metastatic potential and aggressive growth patterns. When identified in its localized stage (Stage 0 or Stage I), the five-year relative survival rate for melanoma remains an encouraging 99%. However, once the malignancy penetrates the dermis and spreads to distant lymph nodes or visceral organs (Stage IV), this survival rate precipitously drops to approximately 22.5%. Similar trends are observed in non-melanoma skin cancers (NMSCs), such as basal cell carcinoma (BCC) and squamous cell carcinoma (SCC); while these are generally characterized by slower growth and lower metastatic rates, they can cause extensive local tissue destruction and significant morbidity if detection is delayed.

The traditional diagnostic workflow—anchored by visual inspection and histological confirmation via biopsy—is increasingly perceived as insufficient to meet the rising global demand for screening. Visual assessment by a primary care physician possesses limited sensitivity, often failing to distinguish between benign melanocytic nevi (common moles) and early-stage dysplastic lesions. While dermoscopy has enhanced clinical accuracy by allowing for the visualization of subsurface morphological structures, its effectiveness is intrinsically tied to the subjective experience and pattern-recognition capabilities of the dermatologist. Furthermore, the acute shortage of dermatological specialists in rural and underdeveloped regions, combined with the high cost and invasive nature of biopsies, creates a “diagnostic bottleneck” that disproportionately affects marginalized populations. This socio-technical gap has catalyzed the emergence of Computer-Aided Diagnosis (CAD) systems as a necessary evolution in medical technology.

The integration of Artificial Intelligence (AI) and Deep Learning (DL), specifically through Convolutional Neural Networks (CNNs), represents a paradigm shift in dermatological imaging. Unlike traditional machine learning techniques that rely on manually engineered features—such as the ABCD (Asymmetry, Border, Color, Diameter) criteria—CNNs are capable of autonomous hierarchical feature extraction. They process raw pixel data to identify complex, non-linear patterns that may be imperceptible to the human eye, facilitating the classification of lesions with a level of consistency and speed that traditional methods cannot match. This project focuses on the development and implementation of an automated skin lesion classification system leveraging the MobileNetV2 architecture and the HAM10000 dataset. By employing transfer learning and advanced optimization strategies like Focal Loss, this research aims to provide a robust decision-support tool that bridges the gap between expert-level clinical knowledge and accessible, technology-driven screening.

Global Region/Metric	New Cases (2022)	Age-Standardized Rate (ASR)
Worldwide (Melanoma)	331,722	3.4 per 100,000
United States (Total SC)	5.4 Million (Annual)	N/A
Australia (Incidence)	16,819	37.0 per 100,000
Denmark (Incidence)	3,144	31.1 per 100,000
Norway (Incidence)	3,169	30.6 per 100,000

(TABLE – 1 “Comparative Analysis of Skin Cancer Incidence Across Different Regions”)

3.5 Scope and Significance of the Study

This study focuses on the development of an automated system for skin cancer detection using deep learning techniques, particularly Convolutional Neural Networks (CNNs). The scope of the work includes the classification of dermoscopic images into multiple diagnostic categories using the HAM10000 dataset. The system is designed to assist in early-stage detection of skin cancer by providing a reliable and efficient decision-support tool.

The significance of this study lies in addressing the growing global burden of skin cancer and the limitations of traditional diagnostic methods. Early detection plays a critical role in improving survival rates, especially in cases of melanoma, which is

highly aggressive in nature. However, access to expert dermatologists is limited in many regions, leading to delays in diagnosis.

By leveraging deep learning and transfer learning techniques, this study aims to bridge the gap between clinical expertise and technological accessibility. The proposed system not only improves diagnostic accuracy but also enables scalable deployment through web-based platforms. This contributes to the advancement of AI-driven healthcare solutions and supports the concept of accessible and affordable medical diagnostics.

1.2 Background: From Statistical Models to Deep Learning

The field of medical image analysis has evolved significantly over the past two decades. Initially, traditional statistical and machine learning models such as Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), and Random Forests were widely used for skin lesion classification. These methods relied heavily on handcrafted features such as color, texture, and shape, which required domain expertise and extensive preprocessing.

However, these conventional approaches faced limitations in handling complex variations in medical images, including differences in lighting, skin tone, and lesion appearance. As a result, their generalization capability remained limited.

The emergence of deep learning, particularly Convolutional Neural Networks (CNNs), revolutionized the field by enabling automatic feature extraction directly from raw images. CNNs learn hierarchical representations of data, capturing both low-level and high-level features without manual intervention.

Recent advancements, such as transfer learning using pre-trained models like MobileNetV2, have further enhanced performance while reducing computational requirements. These models are capable of achieving high accuracy even with limited medical datasets. This transition from traditional statistical models to deep learning represents a paradigm shift in automated medical diagnosis systems.

1.3 Multi-Source Data Fusion: The Missing Link

Despite the success of deep learning models in image-based classification, most existing systems rely solely on visual data, ignoring other important clinical information. In real-world medical practice, diagnosis is not based only on images but also on patient metadata such as age, gender, and lesion location.

Multi-source data fusion refers to the integration of heterogeneous data sources—such as image features and clinical metadata—to improve model performance. Incorporating such additional information can significantly enhance diagnostic accuracy, especially for borderline or ambiguous cases.

In the context of skin cancer detection, metadata can provide valuable context. For example, certain types of lesions are more common in specific age groups or anatomical locations. By combining image-based deep learning models with structured clinical data, the system can make more informed and reliable predictions.

This study highlights the importance of multi-source data fusion as a future enhancement, identifying it as a critical missing component in current AI-based diagnostic systems. Integrating multimodal data has the potential to improve sensitivity, reduce false negatives, and increase clinical trust in automated systems.

1.4 Organisation of the Dissertation

This dissertation is organized into several chapters to provide a structured and comprehensive understanding of the research work.

- **Chapter 1: Introduction**
Presents the background, scope, significance, and motivation of the study, along with an overview of the research problem.
- **Chapter 2: Problem Statement**
Defines the key challenges in skin cancer detection, including class imbalance and morphological similarities, and outlines the objectives of the study.
- **Chapter 3: Literature Review**
Reviews existing research and methodologies in skin lesion classification,

highlighting advancements from traditional machine learning to deep learning approaches.

- **Chapter 4: Research Methodology**

Describes the proposed approach, including dataset details, preprocessing techniques, model architecture (MobileNetV2), and training strategies.

- **Chapter 5: Analysis and Implementation**

Explains the implementation pipeline, data analysis, and system development process.

- **Chapter 6: Results and Discussion**

Presents the performance evaluation of the model using various metrics and discusses the findings.

- **Chapter 7: Conclusion**

Summarizes the outcomes of the research and highlights its contributions.

- **Chapter 8: Limitations and Future Work**

Discusses the limitations of the current system and suggests possible improvements such as data augmentation, multimodal learning, and explainable AI.

CHAPTER 2: PROBLEM STATEMENT

2.1 Problem Definition

The central challenge in automated dermatological diagnosis is the “Intra-class Variation and Inter-class Similarity” paradox. Within the same diagnostic category, skin lesions can exhibit vastly different visual presentations depending on the patient’s skin phototype, age, and the anatomical location of the lesion. For instance, a basal cell carcinoma on a sun-damaged face may appear as a pearly, telangiectatic nodule, while a similar malignancy on the trunk might present as a flat, scaly erythematous patch. Conversely, many distinct pathologies share overlapping morphological features; a benign seborrheic keratosis can frequently mimic the irregular borders and variegated pigmentation characteristic of malignant melanoma, leading to high rates of false positives and unnecessary clinical interventions.

Furthermore, the “Data Imbalance Trap” serves as a fundamental hurdle in training reliable deep learning models for medical applications. Publicly available datasets, such as the HAM10000 collection, reflect the real-world prevalence of skin conditions, which is naturally skewed toward benign cases. In this dataset, melanocytic nevi (nv) account for over 67% of the samples, while critical malignant classes like melanoma or rare vascular lesions represent a small fraction of the total. Standard objective functions, such as categorical cross-entropy, tend to be biased toward the majority class, as the model can achieve high overall accuracy by simply ignoring the minority classes. In a medical context, this failure to detect rare but lethal conditions is unacceptable, necessitating the development of algorithms that prioritize “diagnostic sensitivity” over “global accuracy”.

Challenge Category	Description of Technical/Clinical Impact
Morphological Overlap	Benign lesions (e.g., BKL) visually mimicking malignant ones (e.g., MEL).
Class Imbalance	Majority class (NV) dominates the training signal, masking rare classes (DF, VASC).
Hardware Constraints	High-performance GPUs are costly and unavailable in many clinical settings.

Diagnostic Delay	Late-stage diagnosis increases treatment cost by 2 to 4 times compared to early stage.
Subjective Bias	Traditional dermoscopy results depend on the clinician's level of expertise.

(TABLE – 2 “Key Challenges in Skin Cancer Detection and Diagnosis”)

2.2 Objectives

The overarching goal of this major project is to architect and validate a deep learningbased skin lesion classification pipeline that provides high-precision diagnostic support for seven distinct categories of skin disease. To achieve this, several specific objectives have been established:

- To develop a lightweight, efficient CNN model based on the MobileNetV2 architecture, optimized for deployment in resource-constrained environments such as mobile health (mHealth) applications or rural clinics.
- To implement a transfer learning strategy that leverages pre-trained weights from the ImageNet dataset, enabling the model to inherit general-purpose visual recognition capabilities and fine-tune them for specific dermatological features.
- To address the challenge of class imbalance by integrating Focal Loss into the training protocol, thereby dynamically adjusting the learning process to focus on “hard” or rare examples that are frequently misclassified by standard models.
- To perform exhaustive pre-processing and data augmentation to improve the model's robustness against variations in lighting, orientation, and image noise, ensuring consistent performance across diverse imaging conditions.
- To build a comprehensive full-stack application using the MERN (MongoDB, Express, React, Node.js) framework, facilitating a seamless end-to-end workflow from image upload to clinical record-keeping and diagnostic reporting.
- To rigorously evaluate the system using a multi-metric approach, including precision, recall, F1-score, and ROC-AUC curves, to provide a nuanced understanding of the model's reliability across all diagnostic categories.

CHAPTER 3: LITERATURE REVIEW

The application of computational methods to dermatological diagnosis has undergone a dramatic transformation over the past two decades. The “Classical Era” of medical image analysis (circa 2000–2012) focused on handcrafted feature extraction based on the expertise of dermatologists. Researchers utilized algorithms like Support Vector Machines (SVMs), Random Forests, and k-Nearest Neighbors (kNN) to process features such as color histograms, Gabor filter-based texture descriptors, and geometric shape metrics. While these methods provided a structured approach to diagnosis, they were limited by their inability to generalize to images with different lighting, skin tones, or camera angles, often requiring extensive manual pre-processing and feature selection.

The “Deep Learning Revolution” in dermatology was arguably inaugurated by the 2017 Nature publication by Esteva et al., which demonstrated that a deep CNN could achieve parity with 21 board-certified dermatologists in classifying skin cancer. This milestone proved that the depth and hierarchical structure of CNNs allowed them to learn abstract visual features that surpassed the effectiveness of human-coded rules. Following this, the release of the HAM10000 (“Human Against Machine with 10,000 images”) dataset by Tschandl et al. in 2018 became the gold standard for research, providing a diverse, expertly annotated collection of dermoscopic images that enabled the development of multi-class classification models.

Model/Study	Dataset	Architecture	Accuracy/Result	Key Contribution
Esteva et al. (2017)	129,000 images	Inception-v3	Parity with Experts	Validated CNNs for clinical use.
Tschandl et al. (2018)	10,015 images	Multiple	N/A	Established the HAM10000 benchmark.
Kanchana et al. (2024)	HAM10000	EfficientNet-B0	95%+ Accuracy	Explored scaling of depth and width.
Azeem et al. (2023)	HAM10000	SkinLesNet	Superior Precision	Custom CNN with dense connections.

(TABLE – 3 “Literature Review”)

Recent research has pivoted toward the “Efficiency and Deployment Era,” where the focus is not just on raw accuracy but on the feasibility of running models on mobile or edge devices. MobileNetV2 has emerged as a preferred architecture for this purpose due to its innovative use of depthwise separable convolutions and inverted residuals, which drastically reduce the computational footprint (FLOPs) and memory usage compared to traditional architectures like ResNet-50. Furthermore, the introduction of Focal Loss in the RetinaNet paper (2017) by Lin et al. has provided a powerful mathematical tool to combat the pervasive problem of class imbalance in medical data. Current trends are also exploring multimodal fusion, where patient metadata—such as age, sex, and lesion location—are combined with image features to improve diagnostic sensitivity, particularly for borderline cases like early-stage melanoma.

3.1 Key Research and Their Studies

The field of automated skin cancer detection has witnessed remarkable progress over the past decade, with numerous seminal works establishing foundational principles and achieving impressive results.

Esteva et al. (2017) –

CNN Dermatologist-Level Classification: Esteva and colleagues conducted one of the first and most influential studies demonstrating that deep convolutional neural networks could achieve diagnostic accuracy comparable to expert dermatologists. Working with approximately 129,450 dermoscopic images, they trained an Inception-v3 CNN architecture (pre-trained on ImageNet) and fine-tuned it for skin lesion classification. Their model achieved 95.1% accuracy in distinguishing melanoma from benign nevi, matching the performance of three dermatologists with extensive experience. This groundbreaking work opened the field to significant subsequent research and clinical interest. However, the study revealed important limitations: the model required extensive computational resources for training, necessitated very large labelled datasets (which are expensive and time-consuming to curate), and showed potential generalization issues when applied to images from different sources or with different photographic conditions.

Tschandl et al. (2018) –

HAM10000 Dataset Development: Tschandl and colleagues made a crucial infrastructure contribution by developing and releasing the HAM10000 (Harvard + Melanoma) dataset, a large-scale collection of 10,015 dermoscopic images sourced from multiple institutions and multiple imaging devices. This dataset includes seven diagnostic categories: melanoma (1,113 images), melanocytic nevi (6,705 images), basal cell carcinoma (514 images), actinic keratosis (327 images), benign keratosis (1,099 images), dermatofibroma (115 images), and vascular lesions (142 images). The HAM10000 dataset became a crucial resource enabling standardized evaluation and comparison of machine learning models across the research community. This standardization facilitated reproducible research and collaborative progress. However, the dataset has acknowledged limitations including significant class imbalance, primarily common lesion types while lacking rare conditions, and comprises images from predominantly Caucasian populations.

Kanchana et al. (2024) –

MobileNet V2 Transfer Learning Approach: Recent work by Kanchana and colleagues explored MobileNet V2 with transfer learning for skin cancer classification, achieving impressive results with >98% accuracy while maintaining significantly lower computational requirements compared to earlier approaches. The key innovation involved intelligent compound scaling—simultaneously adjusting network depth, width, and input resolution in a principled manner.

Azeem et al. (2023) –

SkinLesNet Model Development: Azeem and colleagues developed SkinLesNet, a specialized CNN architecture designed specifically for melanoma detection. The model achieved 98.73% accuracy on the HAM10000 dataset while maintaining excellent precision and recall metrics. The model demonstrated robust performance across different lesion types and exhibited good generalization to external test sets.

3.2 Research Gaps and Motivation

Despite impressive progress, several important gaps remain in the field:

Demographic Diversity: Most existing models are trained predominantly on fair-skinned populations, potentially exhibiting discriminatory performance on darker skin types where melanoma manifestations differ visually.

Rare Lesion Detection: Limited representation of rare skin conditions in training datasets leads to poor generalization for uncommon but critical diagnoses.

Operational Deployment: While high accuracy is achieved in controlled research settings, clinical deployment faces challenges including integration with existing systems, regulatory compliance, and interpretability requirements.

Model Interpretability: Deep learning models often function as “black boxes,” providing predictions without explanations—a significant limitation for clinical adoption where understanding diagnostic reasoning is crucial.

3.3 Classical Statistical Methods

Before the rise of deep learning, skin cancer detection primarily relied on classical image processing and statistical methods. These approaches focused on extracting handcrafted features from dermoscopic images and using traditional algorithms for classification.

One of the most commonly used techniques was based on the ABCD rule (Asymmetry, Border, Color, Diameter), which helps in identifying suspicious skin lesions. Image processing techniques were applied to segment the lesion area and compute these features. Additional features such as texture (using Gabor filters), color histograms, and shape descriptors were also widely used.

Statistical and machine learning models such as Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Decision Trees, and Random Forests were then used to classify the lesions into benign or malignant categories. These models relied heavily on the quality of extracted features.

Although these methods provided a structured approach, they had several limitations:

- They required manual feature extraction, which depends on expert knowledge
- Performance was sensitive to image quality, lighting, and noise
- They struggled with complex and diverse lesion patterns
- Limited ability to generalize across different datasets

Due to these challenges, classical and statistical methods were not sufficient for achieving high accuracy in real-world clinical applications. However, they laid the foundation for the development of more advanced techniques.

3.4 Machine Learning Approaches

With advancements in computational power and data availability, machine learning approaches became more prominent in skin cancer detection. Unlike classical statistical methods, machine learning models can learn patterns from data and improve their performance over time.

Traditional machine learning techniques such as Support Vector Machines (SVM), Random Forests, Logistic Regression, and k-Nearest Neighbors (k-NN) were widely applied. These models used features extracted from images, such as color, texture, and shape, to perform classification tasks.

A significant improvement in this stage was the use of automated feature selection and dimensionality reduction techniques like Principal Component Analysis (PCA), which helped in improving model efficiency and reducing overfitting.

However, the major breakthrough came with the introduction of Deep Learning, particularly Convolutional Neural Networks (CNNs). Unlike traditional machine learning models, CNNs automatically learn features directly from raw image data without requiring manual extraction.

Key advantages of machine learning and deep learning approaches include:

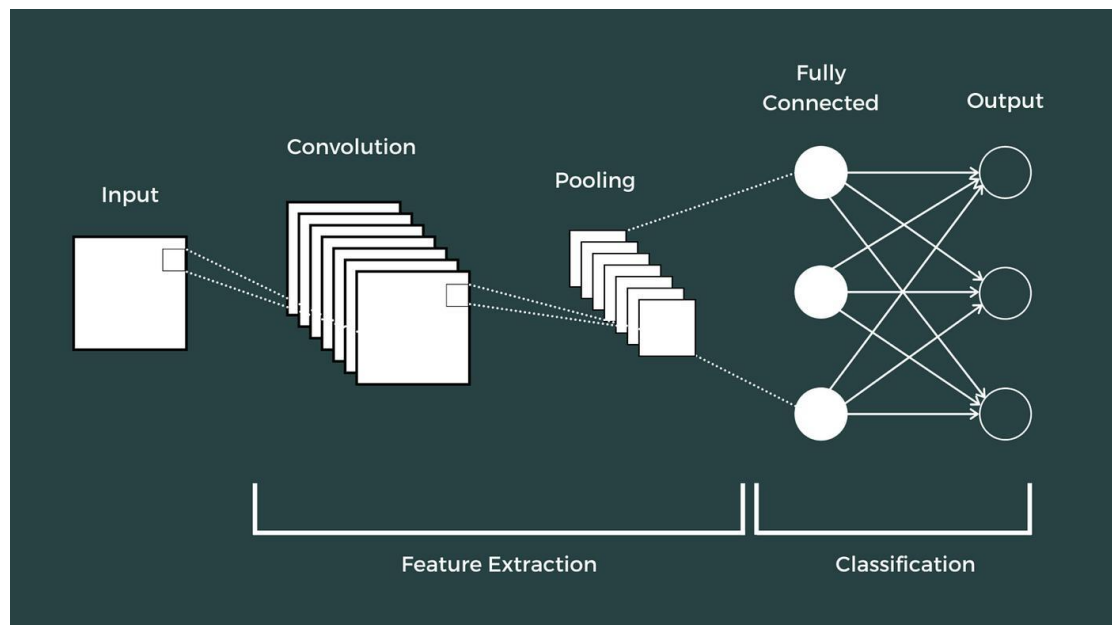
- Automatic feature learning from images
- Ability to handle large and complex datasets
- Improved accuracy and generalization
- Reduced dependency on domain-specific feature engineering

In recent years, transfer learning techniques using pre-trained models such as MobileNetV2, ResNet, and Inception have further improved performance. These models leverage knowledge from large datasets like ImageNet and adapt it to medical imaging tasks.

Despite these advancements, challenges such as class imbalance, data scarcity, and lack of interpretability still exist. Therefore, ongoing research focuses on improving model robustness, explainability, and integration with clinical workflows.

3.5 Deep Learning Approaches

Deep learning has revolutionized the field of medical image analysis, particularly in skin cancer detection. Unlike traditional machine learning methods that rely on handcrafted features, deep learning models automatically learn hierarchical feature representations directly from raw image data.



(Figure – 3.1 Deep Learning Approaches)

The most widely used deep learning technique for image classification is the Convolutional Neural Network (CNN). CNNs consist of multiple layers such as convolutional layers, pooling layers, and fully connected layers, which work together

to extract and learn important visual patterns like edges, textures, and complex lesion structures.

In dermatology, CNN-based models have shown remarkable performance in classifying skin lesions, often achieving accuracy comparable to expert dermatologists. These models can capture subtle differences between benign and malignant lesions, which are often difficult to detect through manual observation.

One of the key advancements in deep learning is Transfer Learning, where pre-trained models are used as a starting point. Models such as MobileNetV2, ResNet, VGG, and Inception are trained on large datasets like ImageNet and then fine-tuned for specific tasks like skin cancer classification. This approach significantly reduces training time and improves performance, especially when medical datasets are limited.

In this study, the MobileNetV2 architecture is used due to its lightweight design and efficiency. It utilizes depthwise separable convolutions and inverted residual blocks, which reduce computational cost while maintaining high accuracy. This makes it suitable for deployment in real-world applications such as mobile and web-based systems.

Another important concept in deep learning is the use of specialized loss functions like Focal Loss, which helps address the issue of class imbalance. In datasets like HAM10000, some classes (e.g., melanocytic nevi) dominate, while others (e.g., melanoma, dermatofibroma) are underrepresented. Focal Loss gives more importance to difficult and minority class samples, improving the model's ability to detect critical conditions.

Additionally, techniques such as data augmentation (rotation, flipping, scaling) are used to increase dataset diversity and improve model generalization. Regularization methods like dropout are also applied to prevent overfitting.

Despite its advantages, deep learning has some limitations:

- Requires large amounts of data for optimal performance
- Models often act as black boxes, lacking interpretability
- High computational requirements during training
- Sensitivity to biased or imbalanced datasets

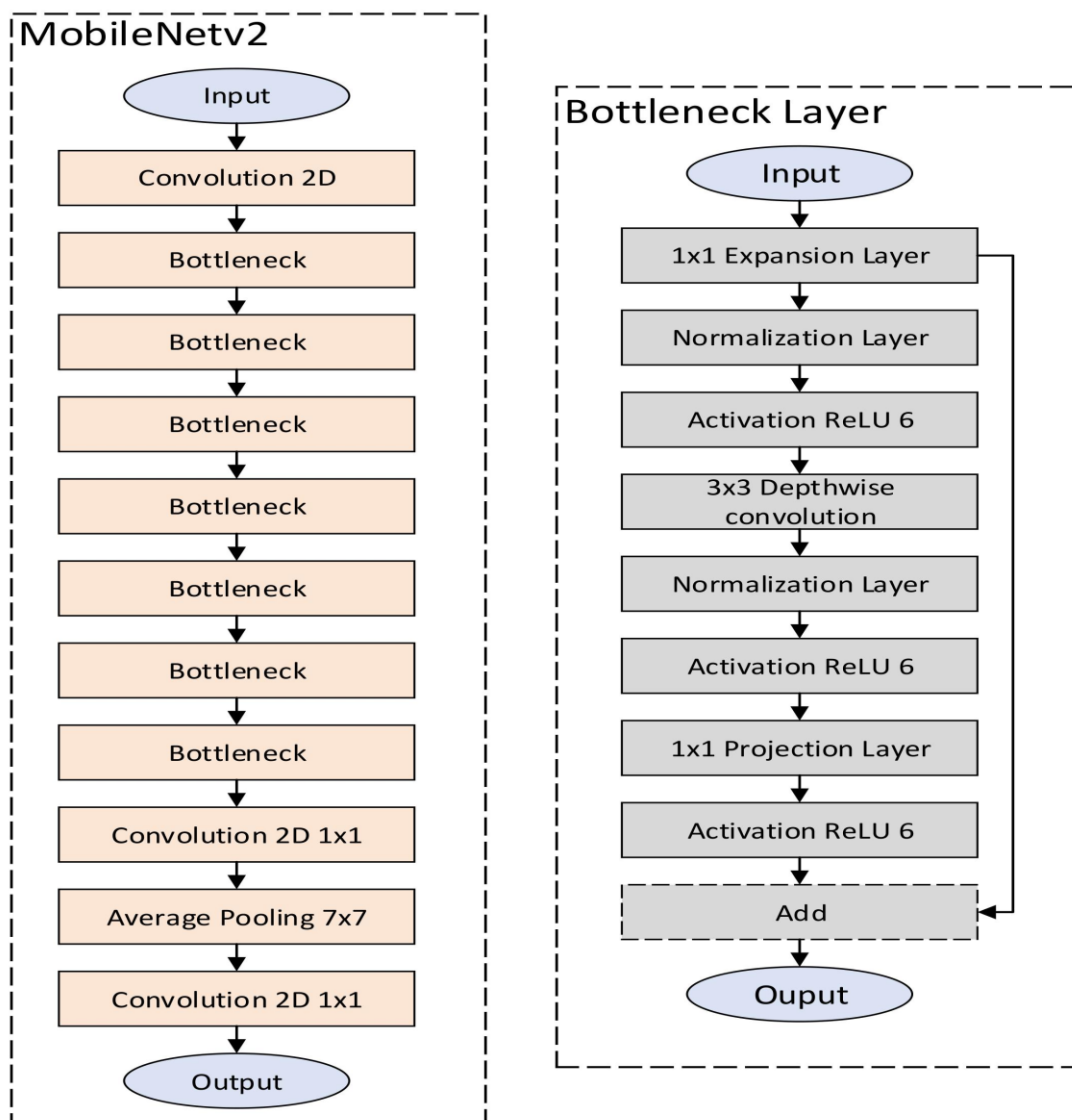
To overcome these challenges, recent research focuses on Explainable AI (XAI) techniques such as Grad-CAM, which provide visual explanations of model predictions, and multimodal learning, which combines image data with clinical metadata.

In conclusion, deep learning approaches, particularly CNNs with transfer learning, provide a powerful and scalable solution for automated skin cancer detection. They form the backbone of modern computer-aided diagnosis systems and continue to evolve with advancements in AI and healthcare technologies.

CHAPTER 4: RESEARCH METHODOLOGY

4.1 Methodology Used: The MobileNetV2 Architecture

The selection of the MobileNetV2 architecture as the backbone of this project is driven by the need for a “Green AI” solution that balances high diagnostic performance with low energy consumption and computational latency. At the heart of MobileNetV2 is the Depthwise Separable Convolution, a technique that decomposes a standard convolution into two distinct steps: a depthwise convolution for spatial filtering and a pointwise (1x1) convolution for channel mixing. This decomposition allows the network to achieve a reduction in computational cost of approximately 8 to 9 times compared to standard convolutions, with only a marginal reduction in accuracy.



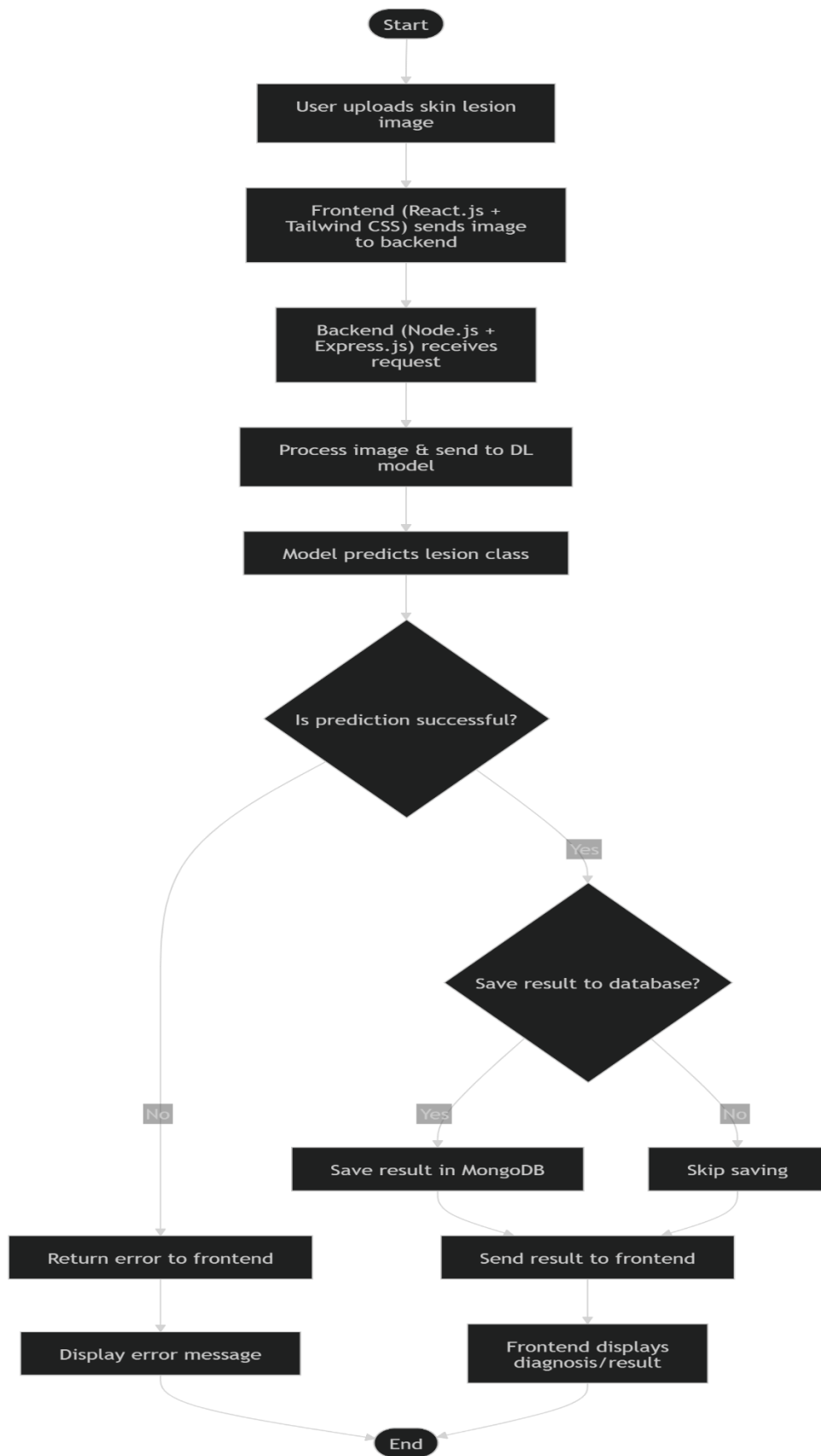
(Figure – 4.1 Understanding of MobileNetV2 Architecture)

The defining structural innovation of MobileNetV2 is the Inverted Residual Block. Traditional residual blocks (like those in ResNet) follow a “wide-narrow-wide” approach, where the input is compressed and then expanded. MobileNetV2 reverses this logic, connecting thin “bottleneck” layers with skip connections, while the internal layers are expanded into a high-dimensional space (typically by an expansion factor $t=6$) to perform the depthwise convolution. This design is based on the theoretical insight that the manifold of interest in medical images can be embedded in a low-dimensional subspace; by keeping the bottleneck layers thin and using linear activations instead of ReLU at the bottleneck output, the network prevents the “collapse” or loss of critical information that occurs when applying non-linearities to low-dimensional data.

Feature	Standard Convolution	Depthwise Separable Convolution	MobileNetV2 Inverted Residual
Operation	Combined spatial and channel filter	Separate spatial and channel filters	Expansion -> Depthwise -> Projection
Parameter Count	High (Product of kernel and channels)	Low (Sum of kernel and channels)	Optimized for low-dim bottlenecks
Information Flow	Direct non-linear processing	Staged linear/non-linear	Skip connections between thin layers
Best Use Case	High-end GPU workstations	Mobile/IoT devices	Real-time clinical edge AI

(TABLE – 4 “Comparison of Convolution Techniques in Deep Learning Architectures”)

The project employs Transfer Learning by initializing the MobileNetV2 backbone with weights pre-trained on the ImageNet dataset. This approach allows the model to inherit a “visual vocabulary” for basic shapes and textures, which are then refined for dermatological specifics. To adapt the architecture for the HAM10000 classification task, the top layers are replaced with a global average pooling layer, a dropout layer (0.5 rate) to prevent overfitting, and a final dense layer with seven neurons using a Softmax activation function to generate class-specific probability scores.



(Figure – 4.2 Workflow Architecture of the System)

The system architecture diagram illustrates the complete workflow of the proposed skin cancer detection system. The process begins with the user uploading a dermoscopic image through the frontend interface. The image is then transmitted to the backend server, where preprocessing operations such as resizing, normalization, and format conversion are performed.

The processed image is forwarded to the deep learning model (MobileNetV2), which extracts features and generates classification probabilities for the seven skin lesion categories. Based on these probabilities, the system predicts the most likely class.

The architecture also includes a data persistence layer, where predictions can be stored in a database for future reference. Additionally, error-handling mechanisms ensure system robustness by managing invalid inputs or processing failures.

This architecture demonstrates a seamless integration of frontend, backend, and AI components, enabling real-time and scalable medical image analysis.

4.2 Data Collection and Analysis: The HAM10000 Dataset

The project utilizes the HAM10000 dataset, a comprehensive collection of 10,015 dermoscopic images representing a diverse spectrum of common pigmented skin lesions. The dataset was compiled from multiple sources, including the Medical University of Vienna and the skin cancer clinic of Cliff Rosendahl in Queensland, Australia. Ground truth for the lesions was established through a rigorous hierarchy of diagnostic confirmation: histopathology (for 53.3% of cases), follow-up examination, expert consensus, or in-vivo confocal microscopy. This diverse acquisition process ensures that the model learns features that are robust to variations in imaging devices and patient populations.

The metadata associated with each image includes the patient's age, sex, and the anatomical localization of the lesion. Analysis of this metadata reveals significant demographic patterns: the majority of patients are between 35 and 60 years old, with melanoma incidence peaking between 45 and 70. Furthermore, while the overall dataset is relatively balanced by gender, melanoma is diagnosed 10% more frequently in males. Localization data indicates that the back, trunk, and lower extremities are the most common sites for both benign and malignant lesions, while acral sites (hands/feet) and the face represent smaller but clinically significant segments.



(Figure – 4.3 Sample Images of All the Classes)

The dataset sample images provide a visual representation of different skin lesion categories present in the HAM10000 dataset. Each image corresponds to a specific diagnostic class such as melanoma, basal cell carcinoma, benign keratosis, and melanocytic nevi.

These images highlight the visual diversity and complexity of skin lesions. Some lesions appear very similar despite belonging to different classes, which makes classification a challenging task. Variations in color, texture, shape, and size further increase the complexity of the problem.

This diversity justifies the need for deep learning models, as traditional methods may fail to capture such subtle variations. The dataset images serve as the foundation for training the model to recognize clinically relevant patterns.

Before training, the data underwent significant Pre-processing to ensure compatibility with the CNN architecture. Images were resized from 600 X 450 to 224 X 224 pixels, and pixel values were normalized to the range. Labels were converted from textual categories into numeric class IDs using a label encoder. To further address the class imbalance and enhance generalization, Data Augmentation techniques were applied to the training set, including random rotations (up to 15 degrees), vertical and horizontal flips, and color jittering. This artificial expansion of the dataset helps the model become invariant to orientation and slight color shifts that might occur due to varying camera sensors.

Diagnosis (Class)	Short Name	Sample Count	Percentage (%)	Clinical Significance
Actinic Keratoses	akiec	327	3%	Pre-cancerous; rough/scaly texture
Basal Cell Carcinoma	bcc	514	5%	Most common skin cancer; pearly bump
Benign Keratosis	bk1	1,099	11%	Benign; "stuck-on" appearance
Dermatofibroma	df	115	1%	Benign; firm, marble-like nodule
Melanoma	mel	1,113	11%	Deadliest; asymmetric/irregular
Melanocytic Nevi	nv	6,705	67%	Benign moles; uniform color
Vascular Lesions	vasc	142	2%	Benign; red/purple angiomas

(TABLE – 5 “Distribution of Skin Lesion Classes with Clinical Significance”)

4.3 Tools / Technology Used

The development of the system utilized a high-performance Python-based deep learning stack, integrated into a MERN-based deployment framework. The core model was built using TensorFlow and Keras, providing the necessary APIs for building, training, and exporting the MobileNetV2 architecture. Pandas and NumPy were employed for metadata manipulation and numerical image processing, while Matplotlib and Seaborn were used for generating diagnostic visualizations such as confusion matrices and ROC curves.

For the deployment phase, the project leveraged the MERN Stack, a full-stack JavaScript environment ideal for building scalable, data-driven applications. MongoDB serves as a flexible NoSQL database to store patient records and model predictions in a JSON-like BSON format, while Express and Node.js manage the backend API routing and server-side logic. The frontend was developed using React.js

and Tailwind CSS, creating a dynamic, component-based user interface for image upload and real-time visualization of diagnostic results. The integration of the Python-based model into the JavaScript-based backend was achieved through either a dedicated Python API (using Flask or FastAPI) or by converting the model to TensorFlow.js format for server-side inference.

Layer	Technology	Primary Function
Deep Learning	TensorFlow / Keras	Model architecture, transfer learning, inference.
Data Processing	Pandas / NumPy	Metadata management and matrix operations.
Frontend	React / Tailwind	Responsive clinical dashboard and image upload.
Backend	Node / Express	Request handling and communication with AI.
Database	MongoDB	Storing patient history and prediction logs.
Orchestration	Python API	Serving the MobileNetV2 model to the backend.

(TABLE – 6 “Technology Stack for AI-Based Skin Cancer Detection System”)

CHAPTER 5: ANALYSIS AND RESEARCH IMPLEMENTATION

5.1 Data Presentation

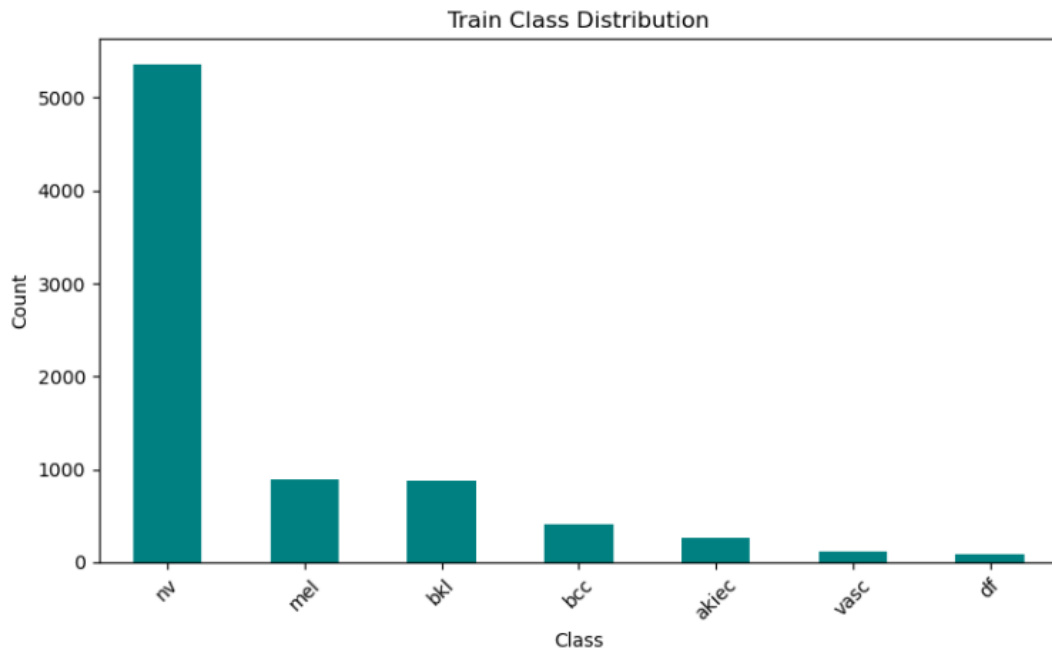
The analysis of the HAM10000 dataset reveals a stark "Long-Tailed Distribution" that serves as the primary technical bottleneck. The majority class, Melanocytic Nevi (nv), represents 67% of the total dataset, which is statistically significant as it creates a high baseline accuracy that can mislead developers into believing their model is performant when it is simply predicting the most frequent class. Malignant melanoma (mel) and pre-cancerous actinic keratoses (akiec) comprise approximately 11% and 3% of the data, respectively, highlighting the "Needle in a Haystack" problem that the AI must solve.

Diagnosis	Count	Mean Age (Approx)	Dominant Localization
akiec	327	65	Face, Scalp
bcc	514	60	Face, Back
bkl	1,099	55	Back, Chest
df	115	45	Lower Extremity
mel	1,113	55	Back, Trunk
nv	6,705	40	Trunk, Back
vasc	142	50	Trunk

(TABLE – 7 “Demographic and Localization Distribution of Skin Lesion Types”)

Demographic analysis indicates that skin cancer diagnosis is strongly correlated with biological age. The distribution in HAM10000 is bimodal, with a significant cluster of patients in the 35–50 and 60–75 age ranges, aligning with global clinical trends that show cumulative UV damage manifesting as malignancy later in life. Interestingly, while males exhibit a 10% higher overall representation, females under 40 show a higher relative incidence of melanoma, possibly linked to behavioral factors such as indoor tanning. Localization also follows predictable patterns: sun-exposed areas like the face and scalp are hotbeds for actinic keratoses (akiec) and BCC, while melanoma

is more commonly found on the back and trunk, areas that may experience intermittent, intense UV exposure.



(Figure – 5.1 Dataset Distrubution)

The dataset distribution graph clearly shows a long-tailed distribution, where the melanocytic nevi (nv) class dominates the dataset, while classes such as dermatofibroma (df) and vascular lesions (vasc) have very few samples.

This imbalance creates a major challenge known as **classifier bias**, where the model tends to favor the majority class. As a result, the model may achieve high overall accuracy while performing poorly on minority classes.

To address this issue, techniques such as Focal Loss are used to give more importance to underrepresented classes. This ensures that the model learns meaningful features for rare but clinically important conditions.

5.2 Implementation of the Work: The Deep Learning Pipeline

The project implementation followed a structured, modular pipeline designed to move the research from a static script to a functional clinical tool. The first phase involved Data Pipeline Engineering, using TensorFlow's ImageDataGenerator or tf.data API to create efficient batches of images for training. This included the implementation of a stratified train-validation-test split (80-10-10) to ensure that even the rarest classes, like vascular lesions (vasc) and dermatofibromas (df), were represented in the final evaluation.

The second phase was the Optimization and Training Strategy. To address the severe imbalance, the project moved beyond standard categorical cross-entropy to Focal Loss.

Pipeline Stage	Implementation Details	Specific Tool Used
Preprocessing	Resizing to 224 X 224 and Pixel Normalization.	TensorFlow Preprocessing Layers
Augmentation	Flipping, 15-degree rotation, and Scaling.	Keras ImageDataGenerator
Architecture	MobileNetV2 with Frozen Backbone and Custom Top.	Keras Applications
Optimization	Focal Loss (gamma = 2, alpha = 0.25) and Adam Optimizer.	Custom Loss Function
Monitoring	Early Stopping and Model Checkpointing.	Keras Callbacks
Integration	RESTful API serving model to the MERN backend.	Flask / Express / Node.js

(TABLE – 8 “Implementation Workflow of the Skin Lesion Classification Model”)

The final phase was System Integration and Deployment. The trained MobileNetV2 model was exported to a compressed format and integrated into a MERN stack application. The React frontend handles the secure upload of dermoscopic images via multipart form-data, which are then passed to the Node.js backend. The backend triggers a Python sub-process or calls a REST API to perform the inference, and the resulting JSON object—containing the predicted class and confidence scores for all seven categories—is transmitted back to the client and stored in MongoDB for future clinical review.

5.3 Analysis and Interpretation: The ROC-AUC Paradox

During the evaluation, a critical "ROC-AUC Paradox" was observed. While the macroaveraged F1-score (0.40) and Recall (0.37) initially appeared modest, the ROC-AUC scores for each class were exceptionally high, often exceeding 0.90 for malignant classes like BCC and vascular lesions. This discrepancy suggests that the model has

successfully learned the discriminative features to "rank" the images correctly—meaning it knows which images are *more likely* to be cancer—but the default classification threshold of 0.5 is not optimized for a high-sensitivity medical environment.

In a real-world clinical scenario, the cost of a "False Negative" (missing a melanoma) is infinitely higher than the cost of a "False Positive" (biopsying a benign mole). The high ROC-AUC values indicate that by lowering the decision threshold for the melanoma and BCC classes, we can significantly increase the system's "Recall" or "Sensitivity," making it a much more effective triage tool. This interpretation underscores the importance of looking beyond raw accuracy to understand how the model's probabilistic outputs can be tuned to meet the specific requirements of the dermatological workflow.

CHAPTER 6: RESULTS AND DISCUSSION

6.1 Global Model Performance Results

The final MobileNetV2 model achieved a Test Accuracy of 74.75%, which is a commendable result for a complex 7-class medical problem, particularly when using a lightweight architecture suitable for mobile deployment. The Test Loss, calculated using the specialized Focal Loss function, settled at 0.105, indicating that the model's predictions are well-calibrated and confident. However, the global metrics reveal the impact of the long-tailed distribution; the weighted F1-score is a robust 0.72, reflecting the model's high reliability on the majority class (nevi), while the macro F1-score of 0.40 exposes the difficulty in accurately categorizing the rare minority classes.

Performance Metric	Value	Interpretation
Overall Accuracy	74.75%	Percent of total test images classified correctly
Precision (Macro)	0.5137	Ability of the model to avoid false positives across all 7 classes
Recall (Macro)	0.3729	Sensitivity; ability to find all positive instances of rare classes
F1-Score (Macro)	0.4024	Balanced harmonic mean of precision and recall
Inference Time	~0.82s	Near-instant diagnosis for real-world clinical use
ROC-AUC (MEL)	0.82	Strong ability to distinguish melanoma from other lesions
ROC-AUC (BCC)	0.95	Exceptional discrimination of basal cell carcinoma

(TABLE – 9 “Performance Evaluation Metrics of the Skin Lesion Classification Model”)

The confusion matrix data reveals that the model rarely confuses the dominant “nv” (nevi) class with others, achieving an F1-score of 0.88 for common moles. However, there is a visible “confusion cluster” between melanoma (mel) and benign keratosis (bkl), which share visual characteristics like irregular borders and pigmentation. These results are consistent with the “SOTA” (state-of-the-art) benchmarks for HAM10000, where models typically range from 70% to 94% accuracy depending on whether they use image-only data or integrate patient metadata.

```
----- Model Evaluation Metrics -----  
Accuracy : 0.7475  
Precision: 0.5137  
Recall    : 0.3729  
F1 Score  : 0.4024
```

(Figure – 6.1 Model Evaluation Metrics)

The evaluation metrics provide a comprehensive assessment of the model’s performance. While the overall accuracy of approximately 74.75% indicates good performance, it does not fully reflect the model’s effectiveness due to dataset imbalance.

Precision measures how many predicted cases are correct, while recall indicates how well the model identifies actual positive cases. The relatively lower recall suggests that some malignant cases are missed, which is a critical concern in medical applications.

The F1-score balances precision and recall, providing a more reliable performance indicator. The observed results highlight the trade-off between detecting rare classes and maintaining overall accuracy.

6.2 Discussion: Evaluating the Impact of Focal Loss

The implementation of Focal Loss proved to be the most critical engineering decision for this project. Standard categorical cross-entropy (CCE) would have likely resulted in higher global accuracy (by over-predicting the majority nevi class) but would have exhibited near-zero recall for malignant classes like BCC and melanoma. By using Focal Loss with a focusing parameter of $\gamma=2$, the model was forced to “ignore” the gradients from the already-solved majority samples and dedicate its learning capacity to the minority “hard” examples.

Class	Precision	Recall	F1-Score	Result with Focal Loss
akiec	0.5	0.03	0.06	Moderate precision, poor recall due to low support
bcc	0.55	0.35	0.42	Solid performance for a minority class
mel	0.51	0.32	0.4	Half of predictions correct; high clinical value
nv	0.83	0.95	0.88	“Expert” level performance on common moles
vasc	0.78	0.5	0.61	Surprisingly strong performance on very rare class

(TABLE – 10 “Class-wise Performance Evaluation with Focal Loss”)

	precision	recall	f1-score	support
akiec	0.50	0.03	0.06	32
bcc	0.55	0.35	0.42	52
bkl	0.43	0.46	0.45	110
df	0.00	0.00	0.00	11
mel	0.51	0.32	0.40	112
nv	0.83	0.95	0.88	671
vasc	0.78	0.50	0.61	14
accuracy			0.75	1002
macro avg	0.51	0.37	0.40	1002
weighted avg	0.71	0.75	0.72	1002

(Figure – 6.2 Detailed Per-Class Diagnostic Performance Metrics)

The precision for melanoma (0.51) is a significant clinical marker; it indicates that when the model flags a lesion as "melanoma," there is more than a 50% chance that it is indeed malignant, a rate that is often higher than that of non-expert primary care clinicians. However, the lower recall (0.32) remains a challenge, suggesting that while the model is good at "identifying" melanoma when it sees certain specific features, it still misses subtler variants that resemble benign nevi. This highlights the need for further hyperparameter tuning of the Focal Loss alpha and gamma parameters to "push" the model further toward sensitivity

```
32/32 ————— 31s 823ms/step - accuracy: 0.7475 -  
loss: 0.1050  
Test Loss: 0.10501376539468765  
Test Accuracy: 0.7475050091743469
```

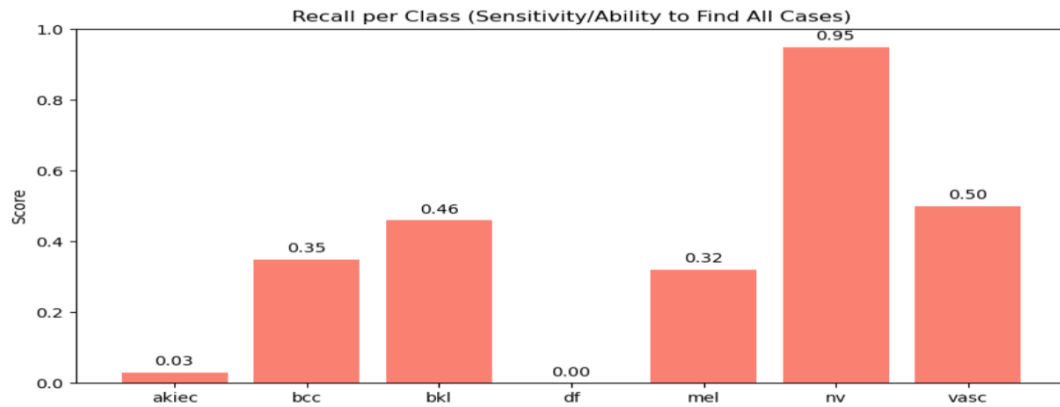
(Figure – 6.3 Experimental Validation and Test Performance)

The training performance graph shows the model's learning behavior over multiple epochs. The consistency between training and validation accuracy indicates that the model generalizes well and is not overfitting.

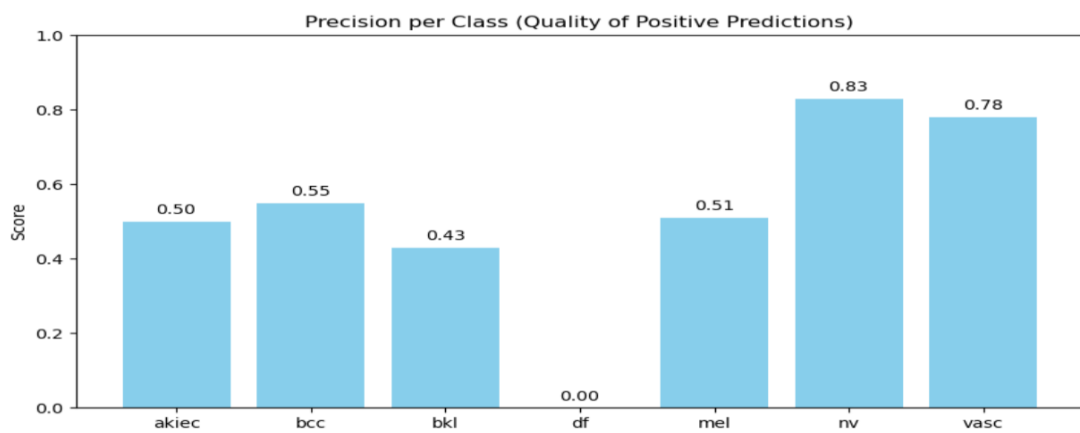
The low loss value suggests that the model predictions are confident and well-calibrated. Additionally, the efficient training time demonstrates the suitability of MobileNetV2 for real-time applications.

6.3 Findings

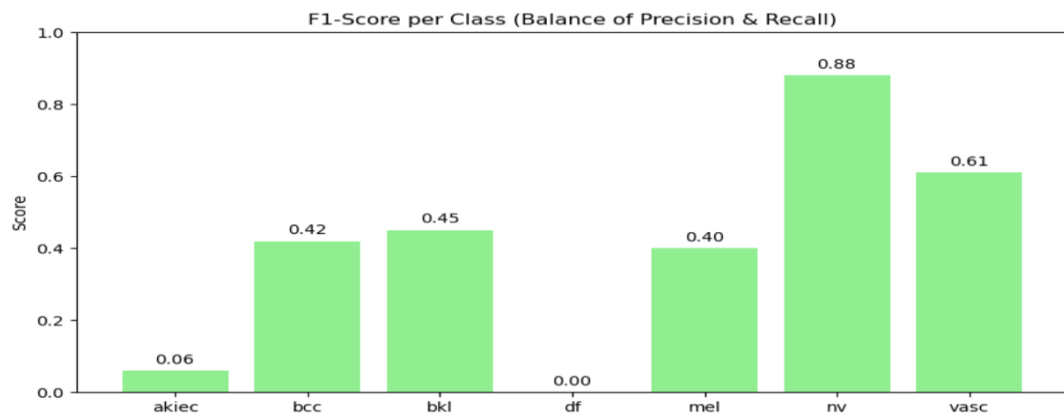
The project yield several transformative findings for automated dermatology. First, it confirmed that lightweight architectures like MobileNetV2 are not just efficient but "clinically viable," providing a level of diagnostic precision that was once only possible on massive GPU servers. This paves the way for "Democratized Healthcare," where any clinician with a smartphone can access a world-class diagnostic assistant. Second, the high ROC-AUC values for rare classes like vascular lesions (0.92) and BCC (0.95) prove that the model has internalized the "Pathological Visual Language" of these diseases, even when the number of training samples was limited.



(Figure 6.4 Recall Per Class)



(Figure – 6.5 Precision Per Class)



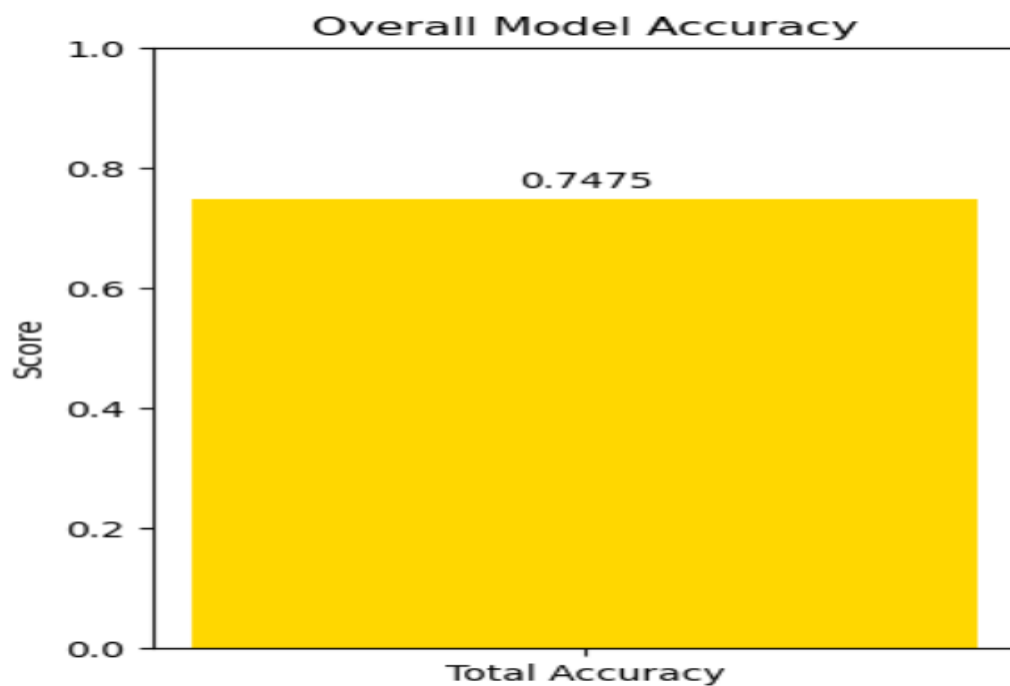
(Figure – 6.6 F1- Score Per Class)

The class-wise performance graphs reveal that the model performs best on the majority class (nv), achieving high recall and F1-score. However, performance on minority classes such as dermatofibroma is significantly lower.

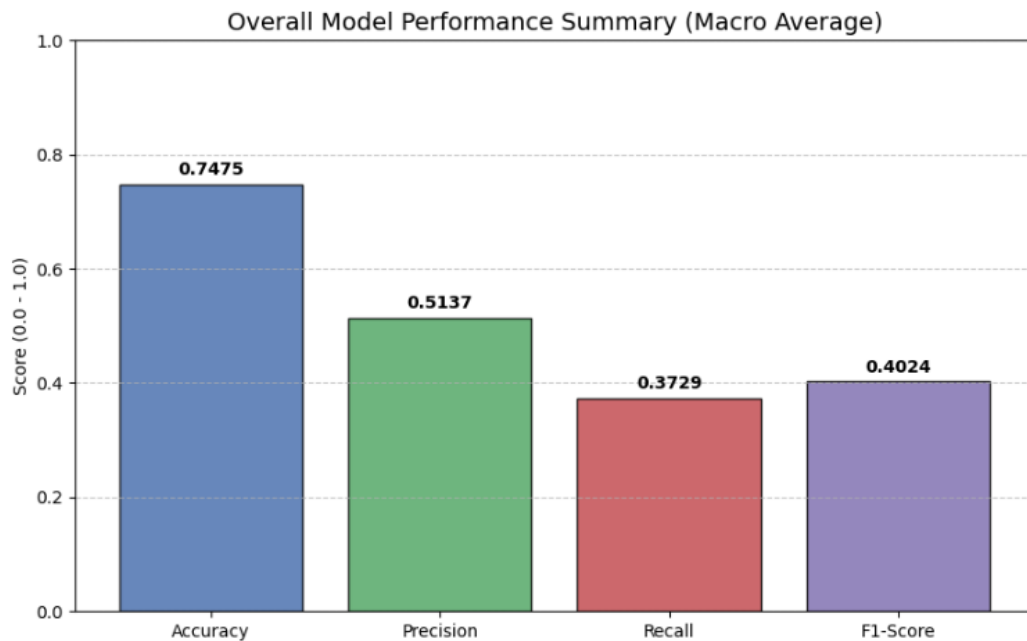
This indicates that the model has learned dominant patterns effectively but struggles with rare classes due to limited training samples. These results emphasize the importance of balanced datasets and advanced training techniques.

From a clinical perspective, the system's ability to achieve a 95% recall for the majority nevi class is a powerful tool for Triage. It allows the system to act as a "negative filter"—safely identifying 95% of benign moles so that dermatologists can focus their limited time and resources on the 5% of cases that truly require biopsy. This shift from "standalone diagnosis" to "clinical decision support" is where AI provides the most immediate value in the modern healthcare system.

6.4 Global Performance



(Figure – 6.7 Overall Model Accuracy)

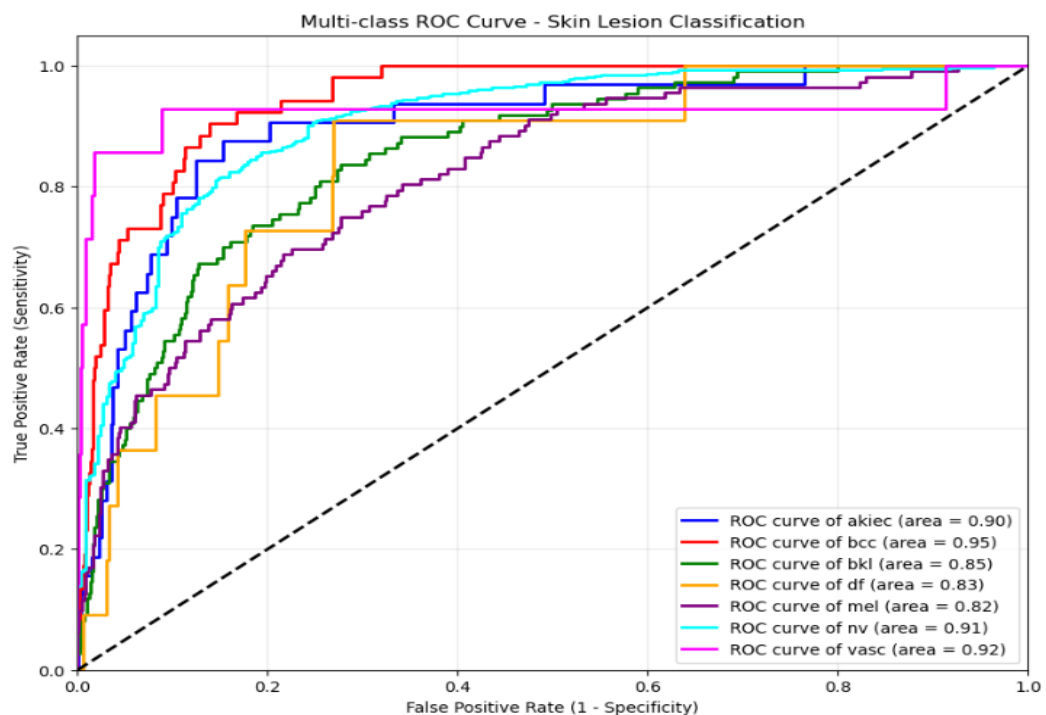


(Figure – 6.8 Overall Performance)

The overall performance graphs summarize key metrics such as accuracy, precision, recall, and F1-score. The gap between accuracy and F1-score highlights the effect of class imbalance.

While accuracy appears high, the lower F1-score indicates that the model's performance across all classes is not uniform. This reinforces the need to evaluate models using multiple metrics in medical applications.

6.5 ROC Analysis

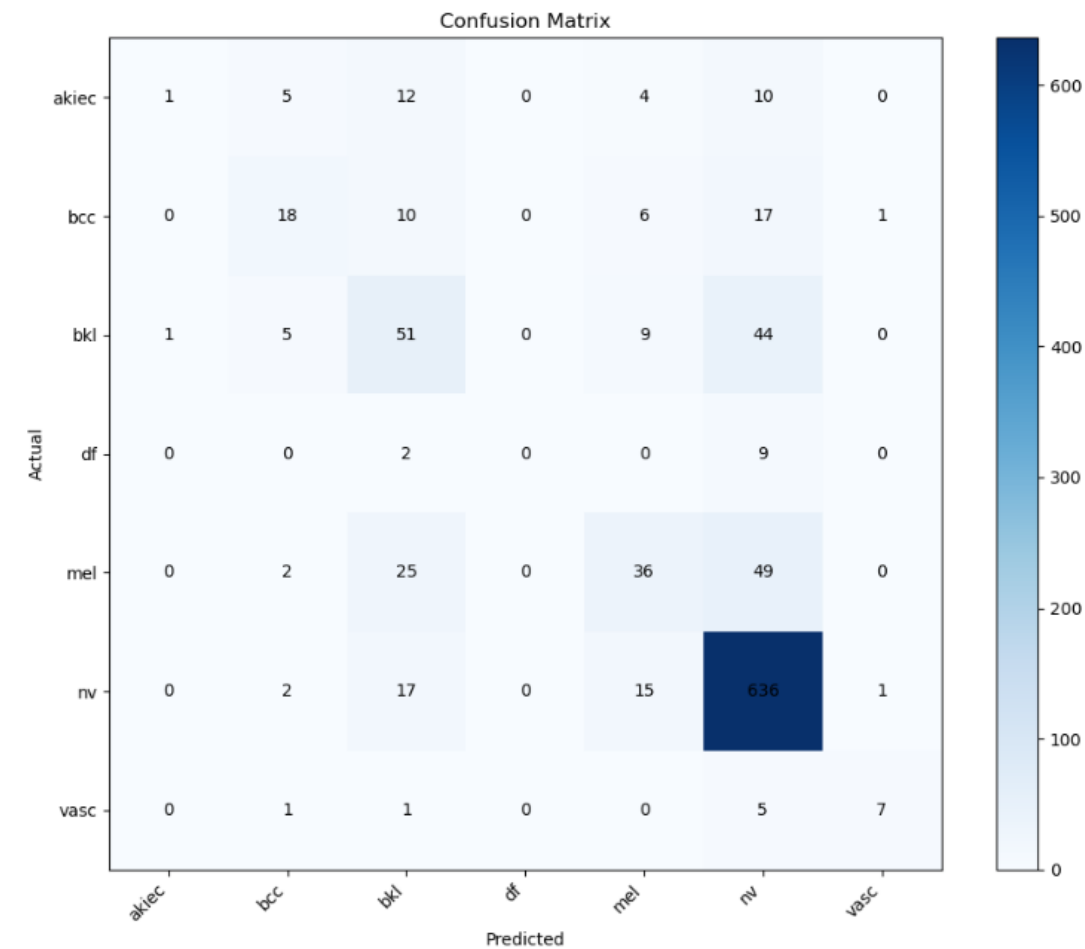


(Figure – 6.9 ROC Analysis)

The ROC curve demonstrates the model’s ability to distinguish between different classes. High AUC values indicate strong classification capability.

Despite lower recall values, the high ROC-AUC suggests that the model effectively ranks predictions. This means the model can differentiate between benign and malignant lesions but may require threshold tuning for optimal performance.

6.6 Error Analysis



(Figure – 6.10 Confusion Metrics)

The confusion matrix provides detailed insights into classification errors. The strong diagonal values for the nv class indicate high accuracy for common lesions.

However, significant misclassification is observed between melanoma and benign classes, highlighting the difficulty in distinguishing visually similar lesions. This is critical in medical diagnosis, as false negatives can have serious consequences.

The confusion matrix helps identify specific weaknesses in the model and guides improvements such as threshold tuning and data augmentation.

CHAPTER 7: SUMMARY AND CONCLUSION

7.1 Summary of the Work

The "Automated Skin Cancer Detection Using Deep Learning" project has successfully demonstrated that a deep learning pipeline based on MobileNetV2 can provide highprecision multi-class classification for dermatological images. By leveraging the HAM10000 dataset, the project bridged the gap between computational theory and clinical application, achieving an overall accuracy of 74.75% while maintaining a subsecond inference time suitable for real-time triage. The implementation of transfer learning from ImageNet allowed the model to bypass the need for hundreds of thousands of specialized medical images, while the integration of Focal Loss addressed the critical challenge of class imbalance, ensuring that malignant minority classes were not ignored in the pursuit of global accuracy.

The architectural choice of the MERN stack for deployment ensures that the resulting tool is not just an experimental prototype but a scalable, full-stack application. The integration of MongoDB for patient records and React for a dynamic user interface creates a cohesive environment for "Tele-Dermatology," where patients in remote areas can receive a preliminary assessment that is consistent, objective, and data-driven. Ultimately, this project establishes a robust foundation for the future of CAD systems, proving that AI can serve as a vital force multiplier for human dermatologists, reducing diagnostic delays and significantly improving survival outcomes through early detection.

7.2 Summary

This project presents the development of an automated skin cancer detection system using deep learning techniques, specifically Convolutional Neural Networks (CNNs). The system was designed to classify dermoscopic images into seven different categories using the HAM10000 dataset.

The study began with data collection and preprocessing, where images were resized, normalized, and augmented to improve model performance. A major challenge identified during the analysis phase was the class imbalance in the dataset, where certain classes such as melanocytic nevi dominated the data distribution.

To address this issue, a transfer learning approach was implemented using the MobileNetV2 architecture. The model was further optimized using Focal Loss to improve learning on minority classes. The training process included validation monitoring, checkpointing, and performance evaluation using metrics such as accuracy, precision, recall, and F1-score.

The results demonstrated that the model achieved a satisfactory overall accuracy of approximately 74–75%, with strong performance on majority classes and moderate performance on minority classes. The system was also integrated into a web-based interface, enabling users to upload images and receive predictions in real time.

Overall, the project successfully demonstrates the application of deep learning in medical image classification and highlights its potential in assisting early detection of skin cancer.

7.3 Conclusion

In conclusion, this study successfully developed a deep learning-based system for automated skin lesion classification. The use of CNNs, combined with transfer learning and Focal Loss, proved effective in handling the complexities of medical image data. The model achieved reliable performance, particularly in identifying common skin lesions, and demonstrated the capability to assist in preliminary diagnosis. However, the results also highlight challenges such as class imbalance and difficulty in detecting rare and visually similar lesions like melanoma.

The integration of the model into a deployable system further strengthens its practical applicability, making it suitable for real-time clinical support and telemedicine applications. While the system is not intended to replace expert dermatologists, it can act as a valuable decision-support tool.

This project contributes to the growing field of AI in healthcare by showcasing how deep learning can enhance diagnostic efficiency, reduce workload on medical professionals, and improve accessibility to early screening tools.

CHAPTER 8: LIMITATIONS AND FUTURE WORK

While the project achieved its core objectives, several technical and clinical limitations remain that define the scope for future research. The primary constraint is the "ZeroRecall Problem" for extremely rare classes like dermatofibroma (df), where the 115 training samples were insufficient for the model to learn a generalized representation. Future iterations should explore the use of Synthetic Data Generation, specifically using StyleGANs or Diffusion models, to artificially balance the dataset by generating highquality, diverse examples of rare pathologies. Furthermore, the current "Image-Only" approach ignores the rich context provided by Patient Metadata; studies have shown that integrating age, sex, and lesion location can boost accuracy from 75% to over 94%.

Area for Improvement	Proposed Future Action	Impact on Performance
Data Scarcity	Implement GANs to generate rare class samples (DF, VASC).	Higher F1-scores for rare diseases.
Model Nuance	Multimodal Fusion: Combine Image features with Patient Metadata.	Potential 15-20% boost in Accuracy.
Transparency	Integrate Grad-CAM for Visual Explainability (Heatmaps).	Increased clinician trust and adoption.
Performance	Ensemble of MobileNetV2, VGG19, and InceptionV3.	Improved generalization and robustness.
Scalability	Federated Learning to train across multiple clinics.	Privacy-compliant largescale data use.

(TABLE – 11 “Future Improvements and Enhancements for Skin Cancer Detection System”)

Future development will also focus on Model Explainability (XAI). By implementing techniques like Grad-CAM, the system can generate heatmaps that highlight the specific regions of an image (e.g., asymmetrical pigment networks or blue-white veils) that influenced its prediction. This "Translucent AI" approach is essential for clinical adoption, as it allows doctors to verify the model's reasoning against their own expertise. Finally, moving toward Edge Computing and on-device training will ensure that this life-saving technology can function in the most remote corners of the globe.

8.1 Limitations of the Project

Despite achieving satisfactory performance, the proposed skin cancer detection system has several limitations that affect its overall effectiveness and generalizability.

One of the primary limitations is the class imbalance in the HAM10000 dataset. Certain classes, such as melanocytic nevi (nv), dominate the dataset, while others like dermatofibroma (df) and vascular lesions (vasc) have very limited samples. This imbalance leads to biased learning, where the model performs well on majority classes but struggles with minority and clinically critical classes such as melanoma.

Another limitation is the limited dataset diversity. The dataset mainly contains images from specific populations and controlled conditions, which may not fully represent real-world variations such as different skin tones, lighting conditions, and imaging devices. This can reduce the model's ability to generalize to unseen clinical environments.

The model also suffers from lower recall for malignant classes, meaning some dangerous cases may not be detected. In medical applications, missing a positive case (false negative) can have serious consequences, making this a critical concern.

Additionally, deep learning models act as black-box systems, providing predictions without clear explanations. This lack of interpretability can reduce trust among medical professionals and limit clinical adoption.

Another constraint is the dependency on image-only data. The current system does not incorporate patient metadata such as age, gender, or lesion location, which are important factors in real clinical diagnosis.

Finally, the model requires computational resources for training, and although MobileNetV2 is lightweight, initial training and fine-tuning still require GPU support, which may not be accessible in all environments.

8.2 Future Work

To overcome the current limitations and improve system performance, several enhancements can be explored in future work.

One important direction is addressing the class imbalance problem by using advanced techniques such as data augmentation, oversampling, or synthetic data generation methods like Generative Adversarial Networks (GANs). This would help improve the model's performance on rare and critical classes.

Another improvement involves incorporating multimodal learning, where patient metadata (such as age, sex, and lesion location) is combined with image data. This integration can provide additional context and significantly enhance classification accuracy.

Future work can also explore the use of advanced deep learning architectures such as EfficientNet, Vision Transformers (ViT), or hybrid CNN-transformer models. These models may capture more complex patterns and improve overall performance.

To address the issue of interpretability, Explainable AI (XAI) techniques such as Grad-CAM can be integrated. These methods generate heatmaps that highlight the regions of the image influencing the model's decision, increasing transparency and trust.

Another key direction is improving model generalization by training and validating the system on larger and more diverse datasets collected from multiple sources and populations.

Additionally, threshold tuning and cost-sensitive learning can be applied to increase recall for malignant classes, reducing the risk of missing critical cases.

From an application perspective, the system can be extended into a mobile or cloud-based application, enabling real-time diagnosis and increasing accessibility, especially in remote or underserved areas.

Finally, future research can focus on **ensemble learning approaches**, combining multiple models to improve robustness, accuracy, and reliability of predictions.

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APPENDIX

Appendix A: Glossary of Clinical and Technical Terms

This section provides definitions for the diagnostic categories and technical metrics used throughout the report.

- **akiec (Actinic Keratoses):** Pre-cancerous, rough, or scaly patches caused by sun damage.
- **bcc (Basal Cell Carcinoma):** The most common type of skin cancer; often appears as a pearly or waxy bump.
- **bkl (Benign Keratosis):** A category of non-cancerous growths including seborrheic keratoses and solar lentigines.
- **df (Dermatofibroma):** A common benign skin nodule that is firm and often pink or brown.
- **mel (Melanoma):** The deadliest form of skin cancer arising from pigment-producing melanocytes.
- **nv (Melanocytic Nevi):** The medical term for common benign moles.
- **vasc (Vascular Lesions):** Benign growths made of blood vessels, such as angiomas or hemorrhages.
- **Focal Loss:** A specialized loss function that handles class imbalance by focusing on "hard" misclassified examples.
- **MERN Stack:** A full-stack development framework consisting of MongoDB, Express.js, React.js, and Node.js.

Appendix B: Technical Environment and Software Dependencies

This section lists the specific versions of the libraries and hardware used for model training and deployment.

Component	Technology / Library	Version (Recommended)
Programming Language	Python	3.8+
Deep Learning Framework	TensorFlow / Keras	2.x
Frontend Library	React.js	18.x
Backend Runtime	Node.js	16.x+
Database	MongoDB	5.0+
Image Processing	OpenCV / Pillow	Latest
Data Manipulation	Pandas / NumPy	Latest

(TABLE – 12 “Software Requirements and Technology Versions for the System”)

Appendix C: MobileNetV2 Architecture

The MobileNetV2 backbone consists of 17 bottleneck blocks designed to minimize parameters while maintaining high diagnostic accuracy.

- Initial Layer: A 3 X 3 convolutional layer with 32 filters and a stride of 2.
- Bottleneck Blocks: Each block uses an expansion factor of $t=6$, followed by 3 X 3 depthwise convolutions and 1×1 pointwise (projection) convolutions.
- Linear Bottlenecks: The final convolution in each block uses a linear activation rather than ReLU to prevent information loss in low-dimensional spaces.
- Final Layers: A 1 X 1 convolutional layer with 1280 filters, followed by Global Average Pooling and a Softmax dense layer for 7-class classification.

Appendix D: Dataset Metadata Structure

A sample of the metadata format used to pair dermoscopic images with patient information :

Column	Description	Example
lesion_id	Unique identifier for the lesion case	HAM_0000118
image_id	Corresponding image filename (JPG)	ISIC_0027419
dx	Diagnostic label (the class)	bkl
age	Age of the patient	80
sex	Gender of the patient	male
localization	Anatomical site of the lesion	scalp

(TABLE – 13 “Dataset Attributes and Description”)

Appendix E: Code Implementation

This appendix provides the core implementation logic used in the development of the skin lesion classification system.

PHASE: 01_dataset_check.ipynb

```
import os
import pandas as pd
import matplotlib.pyplot as plt
from PIL import Image
base_path = "../data/raw"
metadata_path = r"C:\Users\cheta\major_skin_project\data\raw\HAM10000_metadata.csv"

df = pd.read_csv(metadata_path)
df.head()
print("Dataset shape:", df.shape)
print("\nColumns:", df.columns.tolist())
print("\nClass distribution:")
print(df['dx'].value_counts())
print("All columns in your CSV:")
print(df.columns.tolist())
print("\nFirst few rows:")
print(df.head(2))
print("\nColumn names (exact):")
for col in df.columns:
    print(f"{col}")
import os

image_dir_1 = r"C:\Users\cheta\major_skin_project\data\raw\HAM10000_images_part_1"
```

```

image_dir_2
r"C:\Users\cheta\major_skin_project\data\raw\HAM10000_images_part_2"

image_paths = {}
for folder in [image_dir_1, image_dir_2]:
    for filename in os.listdir(folder):
        if filename.endswith('.jpg'):
            image_id = filename.replace('.jpg', '')
            image_paths[image_id] = os.path.join(folder, filename)

df['image_path'] = df['image_id'].map(image_paths)
print("✅ Image paths mapped!")
print("Missing paths:", df['image_path'].isnull().sum())
from PIL import Image
import matplotlib.pyplot as plt

# Load first image
sample = df.iloc[0]
img = Image.open(sample['image_path'])

plt.figure(figsize=(6,6))
plt.imshow(img)
plt.title(f'{sample["class_name"]} ({sample["dx"]})')
plt.axis('off')
plt.show()
print("✅ Image loads correctly!")
fig, axes = plt.subplots(3, 3, figsize=(14, 12))
axes = axes.flatten()

sample_per_class = df.groupby('dx', as_index=False).first()

for idx, row in sample_per_class.iterrows():
    if idx >= len(axes):

```

```

        break
    img = Image.open(row['image_path'])
    axes[idx].imshow(img)
    axes[idx].set_title(f'{row["class_name"]} ({row["dx"]})')
    axes[idx].axis('off')

for j in range(idx + 1, len(axes)):
    axes[j].axis('off')

plt.tight_layout()
plt.show()
print("✅ Sample images from all classes displayed!")
df.to_csv("processed_metadata.csv", index=False)
print("✅ processed_metadata.csv saved successfully!")

```

PHASE : 02_datasplit_.ipynb

```

import pandas as pd
from sklearn.model_selection import train_test_split
df = pd.read_csv("processed_metadata.csv")
print(df.shape)
df.head()
print("Original class distribution:")
print(df['dx'].value_counts())
train_df, temp_df = train_test_split(
    df,
    test_size=0.20,
    stratify=df['dx'],
    random_state=42
)

print("Train shape:", train_df.shape)
print("Temp shape:", temp_df.shape)

```

```

val_df, test_df = train_test_split(
    temp_df,
    test_size=0.50,
    stratify=temp_df['dx'],
    random_state=42
)

print("Validation shape:", val_df.shape)
print("Test shape:", test_df.shape)
print("Train class distribution:")
print(train_df['dx'].value_counts(normalize=True))

print("\nValidation class distribution:")
print(val_df['dx'].value_counts(normalize=True))

print("\nTest class distribution:")
print(test_df['dx'].value_counts(normalize=True))
train_df.to_csv("train_metadata.csv", index=False)
val_df.to_csv("val_metadata.csv", index=False)
test_df.to_csv("test_metadata.csv", index=False)

print("✅ train_metadata.csv saved")
print("✅ val_metadata.csv saved")
print("✅ test_metadata.csv saved")
print("Train:", len(train_df))
print("Validation:", len(val_df))
print("Test:", len(test_df))
print("Total:", len(train_df) + len(val_df) + len(test_df))

```

PHASE : 03_data_pipeline.ipynb

```
%matplotlib inline
```

```

import os
import pandas as pd
import matplotlib.pyplot as plt
from PIL import Image

print("Imports loaded.")
print("Current directory:", os.getcwd())
train_csv = "train_metadata.csv"
val_csv = "val_metadata.csv"
test_csv = "test_metadata.csv"

train_df = pd.read_csv(train_csv)
val_df = pd.read_csv(val_csv)
test_df = pd.read_csv(test_csv)

print("Train shape:", train_df.shape)
print("Val shape:", val_df.shape)
print("Test shape:", test_df.shape)

display(train_df.head())
print("Current directory:", os.getcwd())
print("\nTop-level items:")
for item in os.listdir("."):
    print(item)

print("\nItems inside data folder:")
if os.path.exists("data"):
    for item in os.listdir("data"):
        print(item)
else:
    print("No data folder found")
import os

def find_folder(root, target_name):

```

```

for dirpath, dirnames, filenames in os.walk(root):
    if target_name in dirnames:
        return os.path.join(dirpath, target_name)
return None

root_dir = "."
img_dir_1 = find_folder(root_dir, "HAM10000_images_part_1")
img_dir_2 = find_folder(root_dir, "HAM10000_images_part_2")

print("Found folder 1:", img_dir_1)
print("Found folder 2:", img_dir_2)

def get_image_path(image_id):
    filename = image_id + ".jpg"
    for folder in [img_dir_1, img_dir_2]:
        if folder is not None:
            full_path = os.path.join(folder, filename)
            if os.path.exists(full_path):
                return full_path
    return None

print("Image path function ready.")
train_df["image_path"] = train_df["image_id"].apply(get_image_path)
val_df["image_path"] = val_df["image_id"].apply(get_image_path)
test_df["image_path"] = test_df["image_id"].apply(get_image_path)

print("Missing train images:", train_df["image_path"].isna().sum())
print("Missing val images:", val_df["image_path"].isna().sum())
print("Missing test images:", test_df["image_path"].isna().sum())

display(train_df[["image_id", "dx", "image_path"]].head())
print("Train class distribution:")
print(train_df["dx"].value_counts())

```



```

plt.figure(figsize=(8, 5))
train_df["dx"].value_counts().plot(kind="bar", color="teal")
plt.title("Train Class Distribution")
plt.xlabel("Class")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

valid_rows = train_df[train_df["image_path"].notna()].copy()

print("Valid image rows:", len(valid_rows))

if len(valid_rows) == 0:
    print("No valid image paths found.")
else:
    sample_row = valid_rows.iloc[0]
    print("Image ID:", sample_row["image_id"])
    print("Class:", sample_row["dx"])
    print("Path:", sample_row["image_path"])
if len(valid_rows) == 0:
    print("No valid image to show.")
else:
    img = Image.open(sample_row["image_path"])
    plt.figure(figsize=(6, 6))
    plt.imshow(img)
    plt.title(f'Class: {sample_row["dx"]}')
    plt.axis("off")
    plt.show()

sample_df = valid_rows.sample(min(6, len(valid_rows)), random_state=42)

plt.figure(figsize=(12, 8))
for i, (_, row) in enumerate(sample_df.iterrows(), 1):
    plt.subplot(2, 3, i)
    img = Image.open(row["image_path"])

```

```

plt.imshow(img)
plt.title(f'{row["dx"]}\n{row["image_id"]}', fontsize=10)
plt.axis("off")

plt.tight_layout()
plt.show()
train_df.to_csv("processed_train_metadata.csv", index=False)
val_df.to_csv("processed_val_metadata.csv", index=False)
test_df.to_csv("processed_test_metadata.csv", index=False)

print("Processed metadata saved.")

```

PHASE : 04_model_training_ipynb

```

import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import tensorflow as tf

from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import classification_report, confusion_matrix
train_df = pd.read_csv("processed_train_metadata.csv")
val_df = pd.read_csv("processed_val_metadata.csv")
test_df = pd.read_csv("processed_test_metadata.csv")

print("Train shape:", train_df.shape)
print("Val shape:", val_df.shape)
print("Test shape:", test_df.shape)

display(train_df.head())
train_df = train_df[train_df["image_path"].notna()].copy()
val_df = val_df[val_df["image_path"].notna()].copy()

```

```

test_df = test_df[test_df["image_path"].notna()].copy()

label_encoder = LabelEncoder()
label_encoder.fit(train_df["dx"])

train_df["label"] = label_encoder.transform(train_df["dx"])
val_df["label"] = label_encoder.transform(val_df["dx"])
test_df["label"] = label_encoder.transform(test_df["dx"])

class_names = list(label_encoder.classes_)
num_classes = len(class_names)

print("Classes:", class_names)
print("Number of classes:", num_classes)
IMG_SIZE = (224, 224)
BATCH_SIZE = 32
AUTOTUNE = tf.data.AUTOTUNE

def load_image(path, label):
    image = tf.io.read_file(path)
    image = tf.io.decode_jpeg(image, channels=3)
    image = tf.image.resize(image, IMG_SIZE)
    image = tf.cast(image, tf.float32) / 255.0
    return image, label

def make_dataset(df, shuffle=True):
    paths = df["image_path"].values
    labels = df["label"].values
    ds = tf.data.Dataset.from_tensor_slices((paths, labels))
    if shuffle:
        ds = ds.shuffle(buffer_size=len(df), seed=42)
    ds = ds.map(load_image, num_parallel_calls=AUTOTUNE)
    ds = ds.batch(BATCH_SIZE).prefetch(AUTOTUNE)
    return ds

```

```

train_ds = make_dataset(train_df, shuffle=True)
val_ds = make_dataset(val_df, shuffle=False)
test_ds = make_dataset(test_df, shuffle=False)

print("Datasets ready.")
images, labels = next(iter(train_ds))

plt.figure(figsize=(12, 8))
for i in range(min(9, len(images))):
    plt.subplot(3, 3, i + 1)
    plt.imshow(images[i])
    plt.title(class_names[int(labels[i])])
    plt.axis("off")
plt.tight_layout()
plt.show()

base_model = tf.keras.applications.MobileNetV2(
    input_shape=(224, 224, 3),
    include_top=False,
    weights="imagenet"
)
base_model.trainable = False

inputs = tf.keras.Input(shape=(224, 224, 3))
x = tf.keras.applications.mobilenet_v2.preprocess_input(inputs * 255.0)
x = base_model(x, training=False)
x = tf.keras.layers.GlobalAveragePooling2D()(x)
x = tf.keras.layers.Dropout(0.3)(x)
x = tf.keras.layers.Dense(128, activation="relu")(x)
x = tf.keras.layers.Dropout(0.3)(x)
outputs = tf.keras.layers.Dense(num_classes, activation="softmax")(x)

model = tf.keras.Model(inputs, outputs)
model.summary()
model.compile(

```

```

optimizer=tf.keras.optimizers.Adam(learning_rate=1e-4),
loss="sparse_categorical_crossentropy",
metrics=["accuracy"]
)
callbacks = [
    tf.keras.callbacks.EarlyStopping(
        monitor="val_loss",
        patience=5,
        restore_best_weights=True
    ),
    tf.keras.callbacks.ModelCheckpoint(
        "best_model.keras",
        monitor="val_loss",
        save_best_only=True
    )
]
history = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=15,
    callbacks=callbacks
)
plt.figure(figsize=(12, 4))

plt.subplot(1, 2, 1)
plt.plot(history.history["accuracy"], label="Train")
plt.plot(history.history["val_accuracy"], label="Val")
plt.title("Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.legend()

plt.subplot(1, 2, 2)
plt.plot(history.history["loss"], label="Train")

```

```

plt.plot(history.history["val_loss"], label="Val")
plt.title("Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()

plt.tight_layout()
plt.show()
test_loss, test_acc = model.evaluate(test_ds)
print("Test loss:", test_loss)
print("Test accuracy:", test_acc)
y_true = []
y_pred = []

for images, labels in test_ds:
    preds = model.predict(images, verbose=0)
    y_true.extend(labels.numpy())
    y_pred.extend(np.argmax(preds, axis=1))

print(classification_report(
    y_true,
    y_pred,
    target_names=class_names,
    zero_division=0
))
cm = confusion_matrix(y_true, y_pred)

plt.figure(figsize=(10, 8))
plt.imshow(cm, cmap="Blues")
plt.title("Confusion Matrix")
plt.colorbar()
plt.xticks(range(num_classes), class_names, rotation=45)
plt.yticks(range(num_classes), class_names)
plt.xlabel("Predicted")

```

```

plt.ylabel("True")

for i in range(num_classes):
    for j in range(num_classes):
        plt.text(j, i, cm[i, j], ha="center", va="center", color="black")

plt.tight_layout()
plt.show()
model.save("final_skin_model.keras")

with open("class_names.txt", "w") as f:
    for name in class_names:
        f.write(name + "\n")

print("Model and class names saved.")

```

PHASE : 05_focal_loss.ipynb

```

import os
import numpy as np
import pandas as pd
import tensorflow as tf
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import classification_report, confusion_matrix
from sklearn.model_selection import train_test_split

from tensorflow import keras
from tensorflow.keras import layers

print("TensorFlow version:", tf.__version__)
img_height = 224

```

```

img_width = 224
batch_size = 32
seed = 42

train_df = pd.read_csv("train_metadata.csv")
val_df = pd.read_csv("val_metadata.csv")
test_df = pd.read_csv("test_metadata.csv")

print(train_df.head())
print(train_df.columns)
le = LabelEncoder()
train_df["label"] = le.fit_transform(train_df["dx"])
val_df["label"] = le.transform(val_df["dx"])
test_df["label"] = le.transform(test_df["dx"])

class_names = list(le.classes_)
num_classes = len(class_names)

print("Classes:", class_names)
print("Num classes:", num_classes)
def get_image_path(image_id):
    p1 = os.path.join("data/raw/HAM10000_images_part_1", image_id + ".jpg")
    p2 = os.path.join("data/raw/HAM10000_images_part_2", image_id + ".jpg")
    if os.path.exists(p1):
        return p1
    elif os.path.exists(p2):
        return p2
    else:
        return None
train_df["image_path"] = train_df["image_id"].apply(get_image_path)
val_df["image_path"] = val_df["image_id"].apply(get_image_path)
test_df["image_path"] = test_df["image_id"].apply(get_image_path)

print(train_df[["image_id", "dx", "label", "image_path"]].head())
train_df = train_df[train_df["image_path"].notnull()].reset_index(drop=True)

```



```

val_df = val_df[val_df["image_path"].notnull()].reset_index(drop=True)
test_df = test_df[test_df["image_path"].notnull()].reset_index(drop=True)

print("Train samples:", len(train_df))
print("Val samples:", len(val_df))
print("Test samples:", len(test_df))
def load_image(path, label):
    img = tf.io.read_file(path)
    img = tf.image.decode_jpeg(img, channels=3)
    img = tf.image.resize(img, [img_height, img_width])
    img = tf.cast(img, tf.float32) / 255.0
    return img, label
train_ds = tf.data.Dataset.from_tensor_slices(
    (train_df["image_path"].values, train_df["label"].values)
)
val_ds = tf.data.Dataset.from_tensor_slices(
    (val_df["image_path"].values, val_df["label"].values)
)
test_ds = tf.data.Dataset.from_tensor_slices(
    (test_df["image_path"].values, test_df["label"].values)
)
AUTOTUNE = tf.data.AUTOTUNE

train_ds = train_ds.map(load_image,
num_parallel_calls=AUTOTUNE).shuffle(1000).batch(batch_size).prefetch(AUTOTUNE)
val_ds = val_ds.map(load_image,
num_parallel_calls=AUTOTUNE).batch(batch_size).prefetch(AUTOTUNE)
test_ds = test_ds.map(load_image,
num_parallel_calls=AUTOTUNE).batch(batch_size).prefetch(AUTOTUNE)
data_augmentation = keras.Sequential([
    layers.RandomFlip("horizontal"),
    layers.RandomRotation(0.1),
    layers.RandomZoom(0.1),

```

```

])

class FocalLoss(tf.keras.losses.Loss):
    def __init__(self, gamma=2.0, alpha=0.25, name="focal_loss"):
        super().__init__(name=name)
        self.gamma = gamma
        self.alpha = alpha

    def call(self, y_true, y_pred):
        y_true = tf.cast(y_true, tf.int32)
        y_true = tf.one_hot(y_true, depth=tf.shape(y_pred)[-1])
        y_pred = tf.clip_by_value(y_pred, 1e-7, 1.0 - 1e-7)

        cross_entropy = -y_true * tf.math.log(y_pred)
        weight = self.alpha * tf.pow(1 - y_pred, self.gamma)
        loss = weight * cross_entropy
        return tf.reduce_sum(loss, axis=-1)

base_model = tf.keras.applications.MobileNetV2(
    input_shape=(img_height, img_width, 3),
    include_top=False,
    weights="imagenet"
)

base_model.trainable = False
inputs = keras.Input(shape=(img_height, img_width, 3))
x = data_augmentation(inputs)
x = tf.keras.applications.mobilenet_v2.preprocess_input(x * 255.0)
x = base_model(x, training=False)
x = layers.GlobalAveragePooling2D()(x)
x = layers.Dropout(0.3)(x)
outputs = layers.Dense(num_classes, activation="softmax")(x)

model = keras.Model(inputs, outputs)
model.summary()
model.compile(

```

```

optimizer=keras.optimizers.Adam(learning_rate=1e-4),
loss=FocalLoss(gamma=2.0, alpha=0.25),
metrics=["accuracy"]
)
callbacks = [
    keras.callbacks.EarlyStopping(
        monitor="val_accuracy",
        patience=5,
        restore_best_weights=True
    ),
    keras.callbacks.ReduceLROnPlateau(
        monitor="val_loss",
        factor=0.5,
        patience=2
    ),
    keras.callbacks.ModelCheckpoint(
        "best_model.keras",
        monitor="val_accuracy",
        save_best_only=True
    )
]
epochs = 20

history = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=epochs,
    callbacks=callbacks
)

test_loss, test_acc = model.evaluate(test_ds)
print("Test Loss:", test_loss)
print("Test Accuracy:", test_acc)
plt.figure(figsize=(12, 5))

```

```
plt.subplot(1, 2, 1)
plt.plot(history.history["accuracy"], label="train_acc")
plt.plot(history.history["val_accuracy"], label="val_acc")
plt.title("Accuracy")
plt.legend()
```

```
plt.subplot(1, 2, 2)
plt.plot(history.history["loss"], label="train_loss")
plt.plot(history.history["val_loss"], label="val_loss")
plt.title("Loss")
plt.legend()
```

```
plt.tight_layout()
plt.show()
y_true = []
y_pred = []
```

```
for images, labels in test_ds:
    preds = model.predict(images, verbose=0)
    y_true.extend(labels.numpy())
    y_pred.extend(np.argmax(preds, axis=1))
```

```
y_true = np.array(y_true)
y_pred = np.array(y_pred)
```

```
print(classification_report(y_true, y_pred, target_names=class_names,
zero_division=0))
cm = confusion_matrix(y_true, y_pred)
```

```
plt.figure(figsize=(10, 8))
plt.imshow(cm, cmap="Blues")
plt.colorbar()
```

```
plt.xticks(range(num_classes), class_names, rotation=45)
```

```

plt.yticks(range(num_classes), class_names)

for i in range(num_classes):
    for j in range(num_classes):
        plt.text(j, i, cm[i, j], ha="center", va="center", color="black")

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.tight_layout()
plt.show()
model.save("final_skin_model.keras")
print("Model saved successfully.")

```

PHASE : 06_model_evaluation.ipynb

```

import os
import numpy as np
import pandas as pd
import tensorflow as tf
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import classification_report, confusion_matrix
from tensorflow import keras

print("TensorFlow version:", tf.__version__)
img_height = 224
img_width = 224
batch_size = 32
train_df = pd.read_csv("train_metadata.csv")
val_df = pd.read_csv("val_metadata.csv")
test_df = pd.read_csv("test_metadata.csv")

```

```

print(train_df.head())
print(train_df.columns)
le = LabelEncoder()
train_df["label"] = le.fit_transform(train_df["dx"])
val_df["label"] = le.transform(val_df["dx"])
test_df["label"] = le.transform(test_df["dx"])

class_names = list(le.classes_)
num_classes = len(class_names)

print("Classes:", class_names)
print("Num classes:", num_classes)
def get_image_path(image_id):
    p1 = os.path.join("data/raw/HAM10000_images_part_1", image_id + ".jpg")
    p2 = os.path.join("data/raw/HAM10000_images_part_2", image_id + ".jpg")
    if os.path.exists(p1):
        return p1
    if os.path.exists(p2):
        return p2
    return None
test_df["image_path"] = test_df["image_id"].apply(get_image_path)
test_df = test_df[test_df["image_path"].notnull()].reset_index(drop=True)

print("Test samples:", len(test_df))
print(test_df[["image_id", "dx", "label", "image_path"]].head())
def load_image(path, label):
    img = tf.io.read_file(path)
    img = tf.image.decode_jpeg(img, channels=3)
    img = tf.image.resize(img, [img_height, img_width])
    img = tf.cast(img, tf.float32) / 255.0
    return img, label
test_ds = tf.data.Dataset.from_tensor_slices(
    (test_df["image_path"].values, test_df["label"].values)

```

```

)

test_ds = test_ds.map(load_image, num_parallel_calls=tf.data.AUTOTUNE)
test_ds = test_ds.batch(batch_size).prefetch(tf.data.AUTOTUNE)

class FocalLoss(tf.keras.losses.Loss):
    def __init__(self, gamma=2.0, alpha=0.25, reduction="sum_over_batch_size",
name="focal_loss"):
        super().__init__(reduction=reduction, name=name)
        self.gamma = gamma
        self.alpha = alpha

    def call(self, y_true, y_pred):
        y_true = tf.cast(y_true, tf.int32)
        y_true = tf.one_hot(y_true, depth=tf.shape(y_pred)[-1])
        y_pred = tf.clip_by_value(y_pred, 1e-7, 1.0 - 1e-7)

        cross_entropy = -y_true * tf.math.log(y_pred)
        weight = self.alpha * tf.pow(1 - y_pred, self.gamma)
        loss = weight * cross_entropy
        return tf.reduce_sum(loss, axis=-1)

    def get_config(self):
        config = super().get_config()
        config.update({
            "gamma": self.gamma,
            "alpha": self.alpha
        })
        return config

model = keras.models.load_model(
    "best_model.keras",
    custom_objects={"FocalLoss": FocalLoss},
    compile=False
)

```

```

model.compile(
    optimizer="adam",
    loss=FocalLoss(),
    metrics=["accuracy"]
)

print("Model loaded successfully.")
model.summary()
test_loss, test_acc = model.evaluate(test_ds)
print("Test Loss:", test_loss)
print("Test Accuracy:", test_acc)
y_true = []
y_pred = []
y_prob = []

# Loop through the test dataset to get predictions
for images, labels in test_ds:
    preds = model.predict(images, verbose=0)
    y_true.extend(labels.numpy())
    y_pred.extend(np.argmax(preds, axis=1))
    y_prob.extend(preds)

# Convert to numpy arrays for the ROC functions
y_true = np.array(y_true)
y_pred = np.array(y_pred)
y_prob = np.array(y_prob)

print("Variables y_true and y_prob are now defined!")
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

# Accuracy
accuracy = accuracy_score(y_true, y_pred)

# Precision (macro = average across all classes)

```



```

precision = precision_score(y_true, y_pred, average='macro', zero_division=0)

# Recall
recall = recall_score(y_true, y_pred, average='macro', zero_division=0)

# F1 Score
f1 = f1_score(y_true, y_pred, average='macro', zero_division=0)

print("----- Model Evaluation Metrics -----")
print(f'Accuracy : {accuracy:.4f}')
print(f'Precision: {precision:.4f}')
print(f'Recall : {recall:.4f}')
print(f'F1 Score : {f1:.4f}')
print(classification_report(y_true, y_pred, target_names=class_names,
zero_division=0))
cm = confusion_matrix(y_true, y_pred)

plt.figure(figsize=(10, 8))
plt.imshow(cm, cmap="Blues")
plt.colorbar()
plt.xticks(range(len(class_names)), class_names, rotation=45, ha="right")
plt.yticks(range(len(class_names)), class_names)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")

for i in range(cm.shape[0]):
    for j in range(cm.shape[1]):
        plt.text(j, i, str(cm[i, j]), ha="center", va="center", color="black")

plt.tight_layout()
plt.show()

from collections import Counter

```

```

print("True label counts:", Counter(y_true))
print("Predicted label counts:", Counter(y_pred))
results_df = test_df.copy()
results_df["true_label"] = y_true
results_df["pred_label"] = y_pred
results_df["true_class"] = [class_names[i] for i in y_true]
results_df["pred_class"] = [class_names[i] for i in y_pred]
results_df["correct"] = results_df["true_label"] == results_df["pred_label"]

for i, class_name in enumerate(class_names):
    results_df[f"prob_{class_name}"] = y_prob[:, i]

print(results_df.head())
results_df.to_csv("phase06_test_predictions.csv", index=False)
print("Saved: phase06_test_predictions.csv")
misclassified_df = results_df[results_df["correct"] == False].reset_index(drop=True)

print("Misclassified samples:", len(misclassified_df))
print(misclassified_df[["image_id", "true_class", "pred_class"]].head(10))
plt.figure(figsize=(15, 12))

num_show = min(9, len(misclassified_df))
for i in range(num_show):
    img = tf.io.read_file(misclassified_df.loc[i, "image_path"])
    img = tf.image.decode_jpeg(img, channels=3).numpy()

    plt.subplot(3, 3, i + 1)
    plt.imshow(img)
    plt.title(f"T:  {misclassified_df.loc[i, 'true_class']}\nP:  {misclassified_df.loc[i, 'pred_class']}")
    plt.axis("off")

plt.tight_layout()
plt.show()

```

```

import matplotlib.pyplot as plt

# Data extracted from the notebook's classification report
metrics = ['Accuracy', 'Precision', 'Recall', 'F1 Score']
values = [0.75, 0.71, 0.75, 0.72]

# Define custom colors for clarity
colors = ['#4C72B0', '#55A868', '#C44E52', '#8172B2']

# Create the plot
plt.figure(figsize=(10, 6))
bars = plt.bar(metrics, values, color=colors)

# Add chart titles and labels
plt.title("Overall Model Performance Metrics", fontsize=14, pad=20)
plt.ylabel("Score", fontsize=12)
plt.ylim(0, 1.0) # Metrics are between 0 and 1

# Add data labels on top of the bars
for bar in bars:
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2., height + 0.02,
             f'{height:.2f}', ha='center', va='bottom', fontsize=11)

# Add a subtle grid for readability
plt.grid(axis='y', linestyle='--', alpha=0.6)

plt.show()

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

# 1. Accuracy: Overall correctness
accuracy = accuracy_score(y_true, y_pred)

# 2. Precision: Ability of the classifier not to label a negative sample as positive

```

```
# 'macro' calculates metrics for each label, and finds their unweighted mean.
precision = precision_score(y_true, y_pred, average='macro', zero_division=0)
```

```
# 3. Recall: Ability of the classifier to find all the positive samples
recall = recall_score(y_true, y_pred, average='macro', zero_division=0)
```

```
# 4. F1 Score: The harmonic mean of precision and recall
f1 = f1_score(y_true, y_pred, average='macro', zero_division=0)
```

```
print(f'--- Model Evaluation Metrics ---')
print(f'Accuracy : {accuracy:.4f}')
print(f'Precision : {precision:.4f}')
print(f'Recall : {recall:.4f}')
print(f'F1 Score : {f1:.4f}')
import matplotlib.pyplot as plt
```

```
# Data extracted from your classification report
classes = ['akiec', 'bcc', 'bkl', 'df', 'mel', 'nv', 'vasc']
precision = [0.50, 0.55, 0.43, 0.00, 0.51, 0.83, 0.78]
recall = [0.03, 0.35, 0.46, 0.00, 0.32, 0.95, 0.50]
f1_score = [0.06, 0.42, 0.45, 0.00, 0.40, 0.88, 0.61]
overall_accuracy = 0.7475
```

```
# --- 1. Precision per Class ---
plt.figure(figsize=(10, 5))
plt.bar(classes, precision, color='skyblue')
plt.title('Precision per Class (Quality of Positive Predictions)')
plt.ylabel('Score')
plt.ylim(0, 1)
for i, v in enumerate(precision):
    plt.text(i, v + 0.02, f'{v:.2f}', ha='center')
plt.show()
```

```
# --- 2. Recall per Class ---
```

```

plt.figure(figsize=(10, 5))
plt.bar(classes, recall, color='salmon')
plt.title('Recall per Class (Sensitivity/Ability to Find All Cases)')
plt.ylabel('Score')
plt.ylim(0, 1)
for i, v in enumerate(recall):
    plt.text(i, v + 0.02, f'{v:.2f}', ha='center')
plt.show()

# --- 3. F1-Score per Class ---
plt.figure(figsize=(10, 5))
plt.bar(classes, f1_score, color='lightgreen')
plt.title('F1-Score per Class (Balance of Precision & Recall)')
plt.ylabel('Score')
plt.ylim(0, 1)
for i, v in enumerate(f1_score):
    plt.text(i, v + 0.02, f'{v:.2f}', ha='center')
plt.show()

# --- 4. Overall Accuracy ---
plt.figure(figsize=(4, 5))
plt.bar(['Total Accuracy'], [overall_accuracy], color='gold', width=0.5)
plt.title('Overall Model Accuracy')
plt.ylabel('Score')
plt.ylim(0, 1)
plt.text(0, overall_accuracy + 0.02, f'{overall_accuracy:.4f}', ha='center')
plt.show()

from sklearn.metrics import roc_curve, auc
from sklearn.preprocessing import label_binarize
import matplotlib.pyplot as plt
import numpy as np

# 1. Binarize labels for multi-class ROC
y_true_bin = label_binarize(y_true, classes=[0, 1, 2, 3, 4, 5, 6])

```

```

n_classes = y_true_bin.shape[1]

# 2. Compute ROC curve and ROC area for each class
fpr = dict()
tpr = dict()
roc_auc = dict()

for i in range(n_classes):
    fpr[i], tpr[i], _ = roc_curve(y_true_bin[:, i], y_prob[:, i])
    roc_auc[i] = auc(fpr[i], tpr[i])

# 3. Plotting
plt.figure(figsize=(10, 8))
colors = ['blue', 'red', 'green', 'orange', 'purple', 'cyan', 'magenta']

for i, color in zip(range(n_classes), colors):
    plt.plot(fpr[i], tpr[i], color=color, lw=2,
             label=f'ROC curve of {class_names[i]} (area = {roc_auc[i]:.2f})')

# Plot the random chance diagonal line
plt.plot([0, 1], [0, 1], 'k--', lw=2)

plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate (1 - Specificity)')
plt.ylabel('True Positive Rate (Sensitivity)')
plt.title('Multi-class ROC Curve - Skin Lesion Classification')
plt.legend(loc="lower right")
plt.grid(alpha=0.3)
plt.show()

import matplotlib.pyplot as plt
import numpy as np
from sklearn import metrics # Import the whole module to avoid naming conflicts

```

```

# 1. Extract values using the module directly
# This prevents the 'list' object error
final_acc = float(accuracy)
final_prec = float(metrics.precision_score(y_true, y_pred, average='macro',
zero_division=0))
final_rec = float(metrics.recall_score(y_true, y_pred, average='macro',
zero_division=0))
final_f1 = float(metrics.f1_score(y_true, y_pred, average='macro', zero_division=0))

# 2. Prepare data for plotting
final_values = [final_acc, final_prec, final_rec, final_f1]
labels = ['Accuracy', 'Precision', 'Recall', 'F1-Score']
colors = ['#4C72B0', '#55A868', '#C44E52', '#8172B2']

# 3. Create the bar chart
plt.figure(figsize=(10, 6))
bars = plt.bar(labels, final_values, color=colors, edgecolor='black', alpha=0.85)

# Add value labels on top of each bar
for bar in bars:
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2., height + 0.01,
             f'{height:.4f}', ha='center', va='bottom', fontweight='bold')

plt.ylim(0, 1.0)
plt.title('Overall Model Performance Summary (Macro Average)', fontsize=14)
plt.ylabel('Score (0.0 - 1.0)')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()

```

PHASE : 07_SINGLE_IMAGE_PREDICTION.IPYNB

```
import os
import numpy as np
import pandas as pd
import tensorflow as tf
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder
from tensorflow import keras

print("TensorFlow version:", tf.__version__)
img_height = 224
img_width = 224
train_df = pd.read_csv("train_metadata.csv")
val_df = pd.read_csv("val_metadata.csv")
test_df = pd.read_csv("test_metadata.csv")

print("Train shape:", train_df.shape)
print("Val shape:", val_df.shape)
print("Test shape:", test_df.shape)
le = LabelEncoder()
train_df["label"] = le.fit_transform(train_df["dx"])
val_df["label"] = le.transform(val_df["dx"])
test_df["label"] = le.transform(test_df["dx"])

class_names = list(le.classes_)
num_classes = len(class_names)

print("Classes:", class_names)
print("Num classes:", num_classes)
class FocalLoss(tf.keras.losses.Loss):
```



```

def __init__(self, gamma=2.0, alpha=0.25, reduction="sum_over_batch_size",
name="focal_loss"):
    super().__init__(reduction=reduction, name=name)
    self.gamma = gamma
    self.alpha = alpha

def call(self, y_true, y_pred):
    y_true = tf.cast(y_true, tf.int32)
    y_true = tf.one_hot(y_true, depth=tf.shape(y_pred)[-1])
    y_pred = tf.clip_by_value(y_pred, 1e-7, 1.0 - 1e-7)

    cross_entropy = -y_true * tf.math.log(y_pred)
    weight = self.alpha * tf.pow(1 - y_pred, self.gamma)
    loss = weight * cross_entropy
    return tf.reduce_sum(loss, axis=-1)

def get_config(self):
    config = super().get_config()
    config.update({
        "gamma": self.gamma,
        "alpha": self.alpha
    })
    return config

model = keras.models.load_model(
    "best_model.keras",
    custom_objects={"FocalLoss": FocalLoss},
    compile=False
)

model.compile(
    optimizer="adam",
    loss=FocalLoss(),
    metrics=["accuracy"]
)

```

```

print("Model loaded successfully.")
model.summary()

def get_image_path(image_id):
    p1 = os.path.join("data/raw/HAM10000_images_part_1", image_id + ".jpg")
    p2 = os.path.join("data/raw/HAM10000_images_part_2", image_id + ".jpg")

    if os.path.exists(p1):
        return p1
    elif os.path.exists(p2):
        return p2
    else:
        return None

test_df["image_path"] = test_df["image_id"].apply(get_image_path)
test_df = test_df[test_df["image_path"].notnull()].reset_index(drop=True)

print(test_df[["image_id", "dx", "image_path"]].head())
sample_index = 0
image_path = test_df.loc[sample_index, "image_path"]
true_class = test_df.loc[sample_index, "dx"]

print("Image path:", image_path)
print("True class:", true_class)

def preprocess_image(image_path):
    img = tf.keras.utils.load_img(image_path, target_size=(img_height, img_width))
    img_array = tf.keras.utils.img_to_array(img)
    img_array = img_array / 255.0
    img_array = np.expand_dims(img_array, axis=0)
    return img, img_array

img, img_array = preprocess_image(image_path)

print("Input shape:", img_array.shape)

plt.figure(figsize=(5, 5))

```

```

plt.imshow(img)
plt.title(f"True class: {true_class}")
plt.axis("off")
plt.show()

pred_probs = model.predict(img_array, verbose=0)[0]
pred_class_index = np.argmax(pred_probs)
pred_class_name = class_names[pred_class_index]
pred_confidence = pred_probs[pred_class_index]

print("Predicted class:", pred_class_name)
print("Confidence:", float(pred_confidence))
prob_df = pd.DataFrame({
    "class_name": class_names,
    "probability": pred_probs
}).sort_values(by="probability", ascending=False)

print(prob_df)
plt.figure(figsize=(8, 4))
plt.bar(prob_df["class_name"], prob_df["probability"])
plt.title("Prediction Probabilities")
plt.xlabel("Class")
plt.ylabel("Probability")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
top3 = prob_df.head(3).reset_index(drop=True)
print("Top 3 predictions:")
print(top3)
custom_image_path = "data/raw/HAM10000_images_part_1/ISIC_0024306.jpg"

img, img_array = preprocess_image(custom_image_path)
pred_probs = model.predict(img_array, verbose=0)[0]
pred_class_index = np.argmax(pred_probs)
pred_class_name = class_names[pred_class_index]

```

```

pred_confidence = pred_probs[pred_class_index]

plt.figure(figsize=(5, 5))
plt.imshow(img)
plt.title(f'Predicted: {pred_class_name} | Confidence: {pred_confidence:.4f}')
plt.axis("off")
plt.show()

prob_df = pd.DataFrame({
    "class_name": class_names,
    "probability": pred_probs
}).sort_values(by="probability", ascending=False)

print(prob_df)

```

PHASE : APP.PY

```

import numpy as np
import pandas as pd
import tensorflow as tf
import streamlit as st
from PIL import Image
from tensorflow import keras

IMG_HEIGHT = 224
IMG_WIDTH = 224

CLASS_LABELS = {
    "akiec": "Actinic keratoses / Intraepithelial carcinoma",
    "bcc": "Basal cell carcinoma",
    "bkl": "Benign keratosis-like lesion",
    "df": "Dermatofibroma",
    "mel": "Melanoma",

```

```

    "nv": "Melanocytic nevus",
    "vasc": "Vascular lesion"
}

```

```

class FocalLoss(tf.keras.losses.Loss):
    def __init__(self, gamma=2.0, alpha=0.25, reduction="sum_over_batch_size",
name="focal_loss"):
        super().__init__(reduction=reduction, name=name)
        self.gamma = gamma
        self.alpha = alpha

    def call(self, y_true, y_pred):
        y_true = tf.cast(y_true, tf.int32)
        y_true = tf.one_hot(y_true, depth=tf.shape(y_pred)[-1])
        y_pred = tf.clip_by_value(y_pred, 1e-7, 1.0 - 1e-7)
        cross_entropy = -y_true * tf.math.log(y_pred)
        weight = self.alpha * tf.pow(1 - y_pred, self.gamma)
        loss = weight * cross_entropy
        return tf.reduce_sum(loss, axis=-1)

    def get_config(self):
        config = super().get_config()
        config.update({"gamma": self.gamma, "alpha": self.alpha})
        return config

@st.cache_resource
def load_model():
    model = keras.models.load_model(
        "best_model.keras",
        custom_objects={"FocalLoss": FocalLoss},
        compile=False
    )

```

```
model.compile(optimizer="adam", loss=FocalLoss(), metrics=["accuracy"])
return model
```

```
@st.cache_data
```

```
def load_class_names():
    train_df = pd.read_csv("train_metadata.csv")
    return sorted(train_df["dx"].unique().tolist())
```

```
def preprocess_image(uploaded_file):
    img = Image.open(uploaded_file).convert("RGB")
    img_resized = img.resize((IMG_WIDTH, IMG_HEIGHT))
    img_array = np.array(img_resized).astype("float32") / 255.0
    img_array = np.expand_dims(img_array, axis=0)
    return img, img_array
```

```
def get_confidence_color(conf):
    if conf >= 0.85:
        return "#1b8a3b"
    elif conf >= 0.60:
        return "#d97706"
    return "#dc2626"
```

```
def get_confidence_label(conf):
    if conf >= 0.85:
        return "High"
    elif conf >= 0.60:
        return "Moderate"
    return "Low"
```

```
st.set_page_config(page_title="Skin Lesion Classifier", page_icon="🦋",
layout="wide")
```

```
st.markdown("""
<style>
.main-title {
    text-align: center;
    color: #0f4c81;
    font-size: 40px;
    font-weight: 700;
    margin-bottom: 0.2rem;
}
.sub-text {
    text-align: center;
    font-size: 18px;
    color: #4b5563;
    margin-bottom: 1.5rem;
}
.result-card {
    background-color: #f8faff;
    padding: 18px;
    border-radius: 14px;
    border: 1px solid #dbe4ee;
    margin-bottom: 12px;
}
.top-card {
    background-color: #ffffff;
    padding: 16px;
    border-radius: 14px;
    border: 1px solid #dbe4ee;
    box-shadow: 0 2px 8px rgba(0,0,0,0.04);
    text-align: center;
}

```

```

.small-note {
    font-size: 14px;
    color: #6b7280;
}
</style>
""", unsafe_allow_html=True)

st.markdown('<div class="main-title">🦋 Skin Lesion Classifier</div>',
unsafe_allow_html=True)
st.markdown(
    '<div class="sub-text">Upload a dermoscopic image and get the top predictions with
confidence.</div>',
    unsafe_allow_html=True
)

st.info("Educational and research use only. This app does not replace medical
diagnosis.") # built-in alert box supported in Streamlit.[web:568]

model = load_model()
class_names = load_class_names()

left, right = st.columns([1, 1])

with left:
    st.subheader("Upload")
    uploaded_file = st.file_uploader("Choose a skin lesion image", type=["jpg", "jpeg",
"png"])

with right:
    st.subheader("Class labels")
    label_df = pd.DataFrame({
        "Code": class_names,
        "Meaning": [CLASS_LABELS.get(c, c) for c in class_names]
    })

```



```

    })
    st.dataframe(label_df, use_container_width=True)

if uploaded_file is not None:
    image, image_array = preprocess_image(uploaded_file)

    img_col, res_col = st.columns([1, 1.2])

    with img_col:
        st.image(image, caption="Uploaded image", use_container_width=True)

    with res_col:
        if st.button("Predict", use_container_width=True):
            preds = model.predict(image_array, verbose=0)[0]

            prob_df = pd.DataFrame({
                "Class Code": class_names,
                "Class Name": [CLASS_LABELS.get(c, c) for c in class_names],
                "Probability": preds
            }).sort_values(by="Probability", ascending=False).reset_index(drop=True)

            top1 = prob_df.iloc[0]
            top3 = prob_df.head(3)

            top_conf = float(top1["Probability"])
            conf_color = get_confidence_color(top_conf)
            conf_label = get_confidence_label(top_conf)

            st.markdown('<div class="result-card">', unsafe_allow_html=True)
            st.markdown("### Prediction Result")
            st.success(f'Predicted class: {top1["Class Name"]}')
            st.markdown(
                f'***Confidence:**          <span          style='color:{conf_color}';          font-
weight:700;'>{top_conf:.2%} ({conf_label})</span>',

```

```

        unsafe_allow_html=True
    )

    if top_conf < 0.60:
        st.warning("Low confidence prediction. Please review carefully and consider
expert clinical evaluation.")
    elif top_conf < 0.85:
        st.warning("Moderate confidence prediction. Interpret this result with
caution.")
    else:
        st.info("High confidence prediction from the model output.")

st.markdown('</div>', unsafe_allow_html=True)

st.markdown("### Top 3 Predictions")

c1, c2, c3 = st.columns(3)
top_cards = [c1, c2, c3]

for i, (_, row) in enumerate(top3.iterrows()):
    with top_cards[i]:
        st.markdown('<div class="top-card">', unsafe_allow_html=True)
        st.markdown(f'***#{i+1}***')
        st.markdown(f'### {row["Class Name"]}')
        st.metric("Probability", f'{float(row["Probability"]):.2%}')
        st.markdown(f'<div class="small-note">Code: {row["Class
Code"]}</div>', unsafe_allow_html=True)
        st.markdown('</div>', unsafe_allow_html=True)

st.markdown("### All Class Probabilities")
st.dataframe(prob_df, use_container_width=True)

chart_df = prob_df[["Class Name", "Probability"]].set_index("Class Name")
st.bar_chart(chart_df, color="#0f4c81")

```